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TATARI SYSTEMS
Annual Report

Compute and Energy Economics in Emerging and Frontier Markets

Emerging-Market Case Studies | Country Deep Dives | Research Reports

December 2025 - Edited By Nathan Banketa and Tammi Tran

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EXECUTIVE LETTER

Meba Michael

As a Founding Partner and Head of Phase II, Meba boasts experience from the private equity industry across multiple continents and sectors

To Our Partners, & Stakeholders,

This year marked the moment when Tatari Systems crossed the threshold from vision to reality. We began as a startup built on first principles, operating with limited resources, untested assumptions, and an uncompromising belief in the long-term necessity of resilient digital infrastructure. Today, we stand as an operational company. That transition, while measured in commissioning milestones and system uptime, was in truth forged through endurance and resolve.

This year, Tatari Systems moved from idea to action. Starting small, we held firm to one core belief - the need for strong digital systems over time. Now, we're live, a working business. This shift wasn't just about deadlines or stable tech; it came down to persistence, focus, in shared effort. The road to operation wasn't straight or easy. Despite regulatory holdups, supply issues, energy limits, funding hurdles, alongside changing economic conditions, we kept moving forward. Through each obstacle, effort stayed focused on real progress. Buildings don't rise from hope alone. They come from steady work in hard times.

We owe sincere thanks to our employees, interns, alongside operators and partners, plus those who backed us early - guiding us when we were least stable. For startups, trust usually comes well before clear results. Yet this past year, faith turned into action. The systems running today show just how real that promise was. The second half brought tougher conditions. While Bitcoin's price swung wildly, global hashrate shifts squeezed profits - challenging near-term views on digital infrastructure. These strains aren't ignored. Yet they're seen clearly: normal phases within slow tech growth. Cycles shape markets. Systems last far longer. Our plan didn't rely on one market condition. Instead, it focused on staying useful, lean, while lasting.

Looking ahead, our goals go further than just mining. While building on this base, we're getting ready to move into AI computing, alongside other valuable digital projects, that fit our technical skills and future plans. Wherever possible, we'll act but only if core conditions back lasting expansion.

Our forward expansion is informed by the same discipline that carried us through our most difficult periods. Growth, for us, is not reactive. It is deliberate. In fact, recent challenges have reinforced it further. Success for startups isn't shown in smooth times rather, it's proven in endurance. In the months to come, our approach stays grounded steady in action, calm in tone. Not focused on dreams alone, but guided by what's been made. Progress isn't just ahead; it starts from where we stand.

Thanks from all of us to everyone who backed Tatari Systems when results weren't obvious yet. Because you believed, today's working platform exists. Year by year, we'll keep proving worthy of that through action not words.

mebamichael

FOUNDING PARTNER
HEAD OF PHASE II

“THE CONCENTRATION OF ADVANCED AI INFRASTRUCTURE IN A HANDFUL OF COUNTRIES AND COMPANIES RISKS CREATING A NEW AI DIVIDE. WE MUST DECENTRALIZE COMPUTING RESOURCES TO EMPOWER THE GLOBAL SOUTH.”

Deemah Al-Yahya,
Secretary-General, Digital Cooperation Organization

The Digital Co-operation Organization (DCO) is an intergovernmental body founded in 2020 to accelerate digital transformation and promote equitable participation in the global digital economy, especially among emerging and developing nations. With members across MENA, Africa, South Asia, and LATAM, DCO is expanding digital infrastructure, fostering cross-border innovation, and enabling inclusive access to digital skills, platforms, and markets.

Global Trends & Emerging Market Opportunities

In 2025, AI-driven compute demand kept global data-center expansion in overdrive, with developers and capital markets prioritizing “AI-ready” capacity & fueling another year of consolidation and large-scale build commitments. The biggest constraint shifted from capital to infrastructure execution: interconnection queues, substation equipment shortages, and long transmission timelines have become the gating items, making “power-first” site selection and utility partnerships the deciding factor for where capacity actually gets delivered. Those bottlenecks are pushing spillover into emerging and secondary markets where land and permitting are less congested and reliable megawatts can be secured sooner, creating outsized opportunity for countries that can package stable power, cooling headroom, and credible off take into bankable projects.

GLOBAL TRENDS & EMERGING MARKET OPPORTUNITIES

The geography of computing is expanding. While North America, Europe, and East Asia currently host the bulk of data centers and mining farms, emerging markets are rapidly emerging as new frontiers for green computing capacity. In particular, Africa's markets present a compelling opportunity, combining untapped renewable energy with growing digital demand.

Green Data Centers on the Rise: Organizations are redesigning facilities with sustainability in mind, from integrating on-site solar/wind power to advanced cooling and energy management. The market for "green" data centers is booming, projected to triple from ~$48 billion in 2025 to $155.8 billion by 2030 (26.4% CAGR). These investments enable lower carbon emissions and cost savings for power-hungry services like cloud computing and AI

Renewables Powering More of the Grid: Globally, the electricity used to power computation is becoming cleaner. As of 2025, over half (≈52%) of Bitcoin mining's energy comes from renewable or zero-carbon sources – a remarkable jump from ~38% in 2022, as miners seek out cheap hydro, wind, and solar energy. Technology giants are also entering into power purchase agreements for wind and solar farms to offset data center usage. This shift not only cuts emissions but also hedges against fossil fuel price volatility.

Efficiency and Innovation: Across the board, efficiency improvements are helping to temper energy growth. Cloud providers and hyperscalers are deploying cutting-edge cooling (e.g., liquid cooling up to 132 kW/rack in new AI-focused designs) and AI-driven energy optimization to reduce waste. Such measures can yield double-digit percentage reductions in power usage. Nonetheless, absolute energy demand from computing is still climbing due to the "digital boom," particularly from AI model training, streaming, and blockchain validations.

Convergence of Computing Workloads: Notably, some traditionally crypto-focused companies are diversifying into AI to leverage their infrastructure. For example, major Bitcoin miners are repurposing facilities for cloud and AI services during crypto down-cycles. In 2025, several mining firms derived 15–20% of their revenue from AI/HPC (high-performance computing) services. This reflects the 33% annual growth projected in global AI compute demand through 2030. By offering GPU-based computing to tech clients, miners monetize excess capacity and improve utilization. Companies such as TeraWulf are even converting entire sites into 70 MW "AI-ready" data centers to serve enterprise clients. These trends underscore how sustainable compute infrastructure can flexibly support multiple booming sectors – from blockchain to machine learning – for higher overall returns.

Overall: The global industry is moving toward a "sustainable compute" paradigm, where energy efficiency and renewable sourcing are becoming standard practice. This is creating new markets and investment opportunities, particularly in regions that can offer clean power at scale.

RENEWABLE ENERGY SOURCES FOR COMPUTING

Not all green energy is equal when it comes to powering large-scale compute operations. Here we compare major renewable sources – with a focus on hydropower, given its relevance to our startup's strategy – and their suitability for data centers, Bitcoin mines, and AI supercomputers:

Hydropower: Hydroelectric dams provide abundant, stable base load power, which is ideal for 24/7 operations. Many countries with significant hydro resources enjoy very low electricity costs. For instance, Brazil obtains over 80% of its electricity from hydropower, enabling a relatively clean data center industry, though droughts can threaten supply and necessitate reliance on backup fossil-fuel plants. Ethiopia similarly generates ~90% of its power from hydro dams , ensuring a largely carbon-free grid. The advantage of hydro is reliability and scale: once built, a dam can run day and night for decades. The downside is the geographic limitation (availability of dam sites) and the environmental impacts of large reservoirs. In emerging markets with untapped rivers, hydro is a cornerstone for green computing growth.

Solar and Wind: Solar PV and wind turbines have become highly cost-effective power sources, with prices per kWh dropping dramatically over the past decade. They are scalable in most regions (especially Africa's sunny climates and coastal/offshore wind corridors). However, their intermittency is a challenge for round-the-clock data center loads, a problem mitigated by energy storage or hybrid systems. We see more data center operators adopting on-site or off-site solar/wind to supplement grid power. For example, South Africa's largest colocation provider, Teraco (Digital Realty), is building a 120 MW solar farm dedicated to powering its data centers, coupled with battery storage and grid tie-ins for reliability.

Solar is also modular; smaller "solar-plus-storage" systems can power edge data centers or remote mining rigs during peak sun hours. Wind, where available, often complements solar (it is windy at different times) and is used in countries such as Morocco and Egypt to provide additional renewable capacity for industry. In summary, solar/wind are key to scaling green computing, but are typically combined with storage or other sources to ensure 24/7 uptime.

Geothermal: Geothermal energy, which taps the Earth's heat, provides a constant, reliable power source similar to hydropower, though it is available only in certain geographies (e.g., the East African Rift Valley). Kenya is a success story here: it has developed significant geothermal plants, which now supply a large share of its electricity. In fact, Kenya's grid is ~90% renewable-based (geothermal plus hydro). This resource is so reliable that it attracted direct investment in technology: Microsoft partnered with Kenya's KenGen utility to source geothermal energy for new data centers in Kenya.

For compute operators, geothermal can offer the best of both worlds, green and base-load, if you're located above a viable reservoir. Ethiopia and Tanzania also have geothermal potential under development. While not as widely available as solar or wind, geothermal energy is a prized energy source for green compute where available.

Other Renewables: Other clean energy options (e.g., nuclear, biomass) play a minor role in emerging markets' compute plans. Nuclear power provides a stable output but is capital-intensive and faces regulatory hurdles; only South Africa has an operational plant in sub-Saharan Africa. Biomass or waste-to-energy could power smaller installations, but scalability is limited. Hydrogen fuel cells may become a future solution for backup power or off-grid data centers, but that is nascent. For now, hydro, solar, wind, and geothermal form the core of green energy strategies for computing infrastructure, often used in combination to balance reliability & cost.

EMERGING MARKETS: NEXT FRONTIER FOR GREEN COMPUTE

The geography of computing is expanding. While North America, Europe, and East Asia currently host the bulk of data centers and mining farms, emerging markets are rapidly emerging as new frontiers for green computing capacity.

Underpenetrated Digital Infrastructure: Africa today accounts for <1% of global data center capacity despite having ~17% of the world's population. This signals a huge growth runway. Major cloud providers and colocation firms are now eyeing African nations for expansion, as evidenced by recent deals. For example, the U.S. development finance institution DFC invested $300 million in Africa Data Centres (a pan-African operator), and Equinix announced a $390 million investment to build data centers in South Africa. Countries like Nigeria, Kenya, Egypt, and South Africa are beginning to attract significant data center investment, setting the stage for a broader African build-out.

Abundant Clean Energy Reserves: Many emerging economies possess substantial untapped renewable energy resources. In Africa, these often far exceed current domestic demand. For instance, Uganda's electricity grid is 99% renewable (mostly hydro), and Ethiopia and Kenya's grids are around 90% renewables-based thanks to hydro and geothermal projects.

This presents an opportunity to power new server farms with minimal carbon impact. Sun-rich countries in the Sahel and Sahara have gigawatts of solar potential, while coastal regions and the Horn of Africa have strong wind corridors. Emerging market governments are increasingly keen to monetize these resources: providing cheap land and energy deals to tech investors in exchange for infrastructure and job creation. The combination of energy abundance and low costs makes markets like East Africa highly attractive for green compute ventures.

Future Demand and Demographics: By 2050, one-quarter of the world's population will live in Africa. This young, mobile-savvy population is driving rapid growth in internet use, fintech, e-government, and other data-heavy e-government, and other data-heavy services. Simply put, the next billion internet users will largely come from emerging markets.

This translates to soaring demand for cloud computing, digital services, and AI applications delivered locally. Establishing data centers in-region not only captures a share of this growing market but also improves service quality (reduced latency) for local users. Investors recognize that future market growth will be in emerging economies – and securing green energy-powered infrastructure there is a forward-looking bet.

Public-Private Support and Policy: Many governments in emerging markets are actively courting investment in data infrastructure. They see it as a catalyst for broader economic development. Incentives such as tax holidays, special economic zones (SEZs) for IT parks, and public-private partnerships in the energy sector are becoming common. For example, Kenya has launched a Tech SEZ (Konza and initiatives like TRIFIC) to attract data center projects with fast-tracked permits and renewable energy links.

Likewise, Ethiopia's regulators streamlined licensing for foreign computing firms and offered duty-free imports and multi-year tax exemptions for mining and data center equipment. This improving regulatory environment in many African and Asian emerging markets gives investors more confidence that large-scale computing facilities can be built and operated profitably. However, challenges remain (e.g., bureaucratic delays in some countries and the need for clear data protection laws), but the overall trend is toward greater openness and stable, long-term partnerships for infrastructure.

Synergies with Grid Development: An advantage of locating power-intensive projects like data centers in emerging markets is their ability to strengthen local power grids and economies. Data centers serve as steady electricity customers, justifying utility investments in new infrastructure, which in turn benefits local communities.

COUNTRY DEEP DIVE #1

ETHIOPIA

Tatari Systems

ETHIOPIA

Ethiopia in December 2025 represents Tatari's flagship emerging-market opportunity: a convergence of physical infrastructure advantages, regulatory liberalization, and a historically undercapitalized carrier-neutral landscape positioned to become East Africa's compute hub.

KEY METRICS (DECEMBER 2025):

Energy: GERD 1,800 MW operational; 5,150+ MW by end 2026; 15,760 GWh annual (vs. 12,000 GWh demand); electricity $0.035-0.038/kWh (1.5x cheaper than Iceland, 1.2x cheaper than Texas)

Telecom: Ethio Telecom 100M+ subscribers, 73% YoY data growth; Safaricom $138M expansion; dual-carrier partnership leverage

Regulatory: 21 mining operators; permit freeze Aug 2025; government pivot to "AI/HPC" reclassifies energy-compute as "productive" vs. "speculative"

Development Finance: World Bank/IFC mini grid eligibility; $91B African mini grid market by 2030

Infrastructure Gap: 56% population off-grid (25M+ people); 160,000 minigrids needed by 2030 vs. 44,800 on current trajectory

AN AERIAL VIEW OF THE GRAND ETHIOPIAN RENAISSANCE DAM, EXPECTED TO HELP DOUBLE ETHIOPIA'S CURRENT ELECTRICITY PRODUCTION CAPACITY

Strategic Thesis: Ethiopia's emergence as East Africa's compute hub is built on climate advantage (cooling) + hydro base load (stability) + telecom liberalization (partnership leverage) + rural electrification gap (distributed model). Tatari's planned energy-compute architecture uniquely addresses all four vectors.

Structural Compute Advantages:

1. **High-Altitude Cool Climate:** Highland altitudes (7-11,000 ft) reduce data center cooling costs by 30-40% vs. lowland African competitors; GERD-adjacent facilities in Addis (7,726 ft elevation) achieve naturally superior PUE (Power Usage Effectiveness) vs. sea-level competitors
2. **Non-Volatile Hydro Supply:** GERD's 15,760 GWh annual capacity provides baseload power with 92-98% renewable mix; unlike variable solar/wind, hydro stability enables 24/7 HPC workloads & predictable cost s
3. **Large Transmission Corridors:** Post-GERD, Ethiopia exports 480+ MW to Kenya, Sudan, Djibouti, Tanzania; regional power trade creates infrastructure assets for African cloud
4. **Undercapitalized Carrier-Neutral Market:** Only 3 MW existing colocation (Raxio); 100M+ telecom subscribers across Ethio Telecom (incumbent) + Safaricom (disruptor); dual-ISP competition creates partnership leverage
5. **Rural Electrification Imperative:** 25M+ off-grid population; 160k minigrids needed by 2030; distributed energy compute model unavailable in mature markets

Tatari's Positioning: Not "BTC mining company" but "East Africa's distributed energy compute infra provider" combining grid-connected HPC in Addis (targeting hyper scale cloud, AI training) with mini grid-mining in rural zones (rural electrification + development).

This addresses both enterprise and development mandates.

HYDRO PARADOX & SOLUTION

A. The Water Paradox: Abundance Above, Scarcity Below

Ethiopia's water situation creates an apparent paradox that the minigrid model uniquely solves:

Abundance (Highland Rivers):

- **Total Water Resources**: 122 billion cubic meters annually (7th largest in Africa)
- **Highland Rainfall**: 1,500-2,500 mm annually in western highlands (90% of total flows)
- **River Systems**: Blue Nile (originating Ethiopia), Omo, Tekeze—all fed by highlands
- **GERD Capacity**: 74 billion cubic meter reservoir; can store 60% of annual flow

Scarcity (Ground-Level Accessibility):

- **Population Without Access**: 56% lack reliable water access (25M+ people)
- **Transmission Loss**: 35-40% of water lost in distribution from storage to population centers
- **Seasonal Variation**: Dry season (May-Oct) sees 60-70% decline in surface water availability
- **Conflict**: Upstream/downstream water competition (Egypt/Sudan disputes)

The Paradox: Ethiopia has the water abundance to power the region AND to electrify its population, yet both remain constrained by infra (transmission) not supply.

B. Minigrids Solve the Transmission Problem
The minigrid-mining model uniquely addresses the transmission constraint:

Traditional Infrastructure Bottleneck:
GERD (15,760 GWh annually) → National Grid (limited transmission capacity) → [35-40% losses in long-distance transmission] → Rural Communities (25M+ off-grid; only 30-50% of grid capacity available) → Result: Stranded energy + Unelectrified population = Economic waste

Minigrid-Mining Solution:
Local Micro-Hydro (100-500 kW) → Distributed Generation (collocated with communities → [Near-zero transmission loss] → Community Load (80 kW) + Mining Load (70-170 kW seasonal) → Result: No transmission bottleneck + Baseload financing for electrification infrastructure

Economics of Distributed Supply: A 200 kW mini-hydro in rural Oromo Region:

- **Annual Generation**: 1,200 MWh
- **Community Load** (365 days): 600 MWh (80 kW × 24h × 310 days effective utilization)
- **Mining Load** (300 days, seasonal): 600 MWh (70-170 kW avg., varies with season)
- **Annual Revenue**: $80K (mining) + $22.5K (community electricity @ $0.15/kWh)
- **Payback**: 4-5 years; IRR 25-30%
- **Impact**: Electrifies 150-200 households + funds infrastructure for 200+ more households

Key Insight: Water abundance becomes an asset (mini-hydro fuel) rather than a constraint, enabling profit-driven rural electrification.

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Key Insight: Water abundance becomes an asset (mini-hydro fuel) rather than a constraint, enabling profit-driven rural electrification.

THE COMPUTE CASE

A. High-Altitude Advantage: Cooling Economics
Ethiopia's highland geography provides a natural competitive advantage for data center operations unmatched elsewhere in Sub-Saharan Africa:

Altitude & Temperature Profile:
- **Addis Ababa Elevation**: 7,726 ft (2,355 m)—Africa's highest capital city
- **Annual Average Temperature**: 62°F (16.5°C)
- **Seasonal Range**: 50-75°F (10-24°C); extremes
- **Cooling Advantage vs. Competitors**:
  - **Nigeria (Lagos, sea level)**: 78-82°F year-round = high cooling load
  - **Kenya (Nairobi, 5,905 ft)**: 65-75°F = moderate cooling
  - **South Africa (Johannesburg, 5,751 ft)**: 50-75°F = comparable
  - **Ethiopia (Addis, 7,726 ft)**: 50-75°F + consistent throughout country = superior

PUE Impact (Power Usage Effectiveness/Total Power):
- **Baseline (sea-level)**: PUE 1.67 (33% overhead)
- **Advanced Free Cooling (Addis altitude)**: PUE 1.15-1.25 (15-25% overhead)
- **Annual Savings**: 200 MW facility saves 40-50 MW on cooling alone vs. Lagos equivalent

Tatari's Competitive Advantage: Deploy "passive cooling" architecture leveraging natural air circulation at Addis altitudes. Annual cooling costs 30-40% lower than Kenya/South Africa equivalents, creating $8-15M annual savings at 100-150 MW scale.

B. Non-Volatile Hydro Supply: The Baseload Advantage
Unlike solar (6-8 hour daily capacity factor) or wind (35-45% variability), GERD provides consistent baseload:

GERD Hydrology & Stability:
- Annual Inflow Variation: ±8-12% (vs. ±40-60% for wind, ±20-30% for seasonal solar)
- Monthly Generation Curve: Predictable seasonal pattern (wet season Nov-Apr, dry May-Oct)
- Drought Resilience: 74B cubic meter reservoir = 2+ year buffer against droughts
- Generation Reliability: 90-95% capacity factor achievable (vs. 50-60% for wind, 25-35% for solar)

For HPC Workloads:
- **AI Model Training:** Requires 24/7 power stability; hydro enables continuous GPU/TPU utilization
- **Cloud Compute SLAs:** 99.99% uptime commitments feasible with non-volatile baseload
- **Cost Predictability:** Hydro rates locked in PPAs; no variable renewable hour arbitrage risk

ETHIOPIAN HIGHLANDS

Transmission Advantages:
- **Regional Power Trade:** $600M+ annual export revenue flows through transmission corridors
- **Infrastructure Density:** Dense transmission routes from GERD → Addis → Kenya/Sudan/Djibouti
- **Corridor Assets:** Tatari gains leverage in conversations with Ethio Telecom and regional utilities for co-location within transmission infrastructure

C. Transmission: Africa's Cloud Distribution Hub
GERD completion creates a new regional power trade architecture positioning Ethiopia as a distribution hub:

Export Markets & Capacity:

- **Kenya:** 200 MW firm supply (Q1 2026 delivery); critical for Nairobi cloud expansion
- **Sudan:** 100 MW (despite civil war; power trade buffer agreement insulates delivery)
- **Djibouti:** 80 MW capacity agreement; critical for Red Sea/Gulf of Aden logistics hub
- **Tanzania**: 100 MW in negotiations
- **Total Export Potential:** 480+ MW generating $600M+ annual revenue by 2027

For Tatari's Cloud Distribution: Tatari-operated facilities in Addis become a regional hub for cloud distribution:

- Latency Advantage: Addis equidistant from Kenya, Uganda, Tanzania, Sudan—central placement reduces East African latency by 15-25% vs. coastal data centers
- Cost Arbitrage: Procure GERD power at $0.035/kWh; resell cloud services to carriers at premium rates
- Regional Carrier Partnerships: Ethio Telecom (100M subscribers) + Safaricom (10M+ growing) + Vodacom (Tanzania) = customer base without additional infrastructure investment

TRENDS & TARIFF SHOCK

A. GERD Dam: September 2025 Inauguration
The Grand Ethiopian Renaissance Dam represents Africa's largest hydropower facility and signals a fundamental energy system transformation [48][52]:

Current Operational Status (December 2025):

- **Turbines Online**: 7 of 13 installed (1,800 MW generation)
- **Full Capacity Timeline**: All 13 online by end-2026 = 5,150+ MW
- **Annual Generation Potential**: 15,760 GWh (vs. current demand ~12,000 GWh)
- **Stranded Power**: 2,000-3,000 GWh annually wasted (could power 500K+ homes)

Regional Export Architecture [25]:

- **Kenya**: 200 MW (Q1 2026); reducing Nairobi's fossil fuel dependence
- **Sudan**:100MW (despite war; continues via buffer)
- **Djibouti**: 80 MW (Port of Djibouti hub supplies)
- **Tanzania**: 100 MW (negotiations ongoing)

B. December 2025 Tariff Shock: Regulatory Reality
The Ethiopian Electric Utility's 122% tariff increase effective December 2025 signals government's energy policy priority shift [23][30][41]:

Rate Structure Transformation:

- **Previous**: $0.032 USD/kWh (uniform for all industrial users)
- **New (Dec 2025)**: Tiered structure; low-voltage industrial (mining) at 3.09 ETB/kWh ($0.025 USD) + demand charges of 441 ETB/kW ($3.50/kW)

Effective All-In Cost: $0.035-0.038USD(vs. $0.032)

Government Rationale [24][41]:

- Rural Electrification Priority: Government allocating stranded power to 25M+ off-grid population rather than Bitcoin mining
- Grid Capacity Management: Mining was consuming 27% of available capacity by mid-2025; tariff increase discourages mining expansion
- **Long-Term Signaling:** Government messaging shifted from "mining support" to "AI/HPC transition"

Competitive Impact:

Jurisdiction	Electricity Cost	BTC Mining Cost	Stability	Outlook
Ethiopia (Dec 2025)	$0.035-0.038	$2,100-2,250	Firm	Stable post-tariff
Texas (wind)	$0.035-0.042	$2,100-2,500	Stable	Long-term contracts
Iceland (geothermal)	$0.040-0.055	$2,400-3,300	Stable	Limited expansion
Kazakhstan	$0.028-0.035	$1,700-2,100	Volatile	Political risk

Bottom Line: Ethiopia remains globally competitive for mining (top-5 globally) but no longer the overwhelmingly dominant player. Competitive advantage now lies in integrated energy-compute (HPC + mining + grid services), not pure mining arbitrage.

C. Tariff Implications for HPC vs. Mining
The tariff structure creates incentives favoring Tatari's "infrastructure provider" positioning:

Mining Income Tax: 30% (clear classification) Data Processing/AI Tax: Potentially 15-20% (undefined—optimization opportunity) Mixed Operations (mining + HPC + grid services): Undefined—Tatari can argue for blended 20-25% rate

Tax Optimization Strategy: A facility offering "Bitcoin mining (30%), AI training (20%), and grid-stabilization services (20%)" could argue for weighted-average ~23% effective rate rather than flat 30% on mining.

COUNTRY DEEP DIVE #2

GEORGIA

Tatari Systems

GEORGIA

Georgia offers a mature, tax-advantaged jurisdiction positioned as the digital gateway b/n Europe & Asia. However, it is a market defined by extreme seasonality—abundant hydro in summer, deficits in winter—requiring a fundamentally different model.

Systemic Realities:

The "Free Zone" Advantage: Operations within Free Industrial Zones (FIZ) (Tbilisi, Poti, Kutaisi) enjoy 0% Corporate Tax, 0% VAT, and 0% Dividend Tax.

Hydro Seasonality Trap: Unlike GERD (steady base load), Georgia's Enguri Dam relies on seasonal melt. The country has a Summer Surplus (May-Aug) but a severe Winter Deficit (Nov-Mar), often necessitating expensive gas imports or power rationing.

The "Golden Era" Legacy: Georgia was once the #2 miner globally (after China). The infrastructure (substations, fiber) exists, but the grid is now strained by "grey market" mining, in the Abkhazia region, forcing the government to regulate non-FIZ operators.

Geopolitical "Middle Corridor": Georgia is positioning itself as the digital trunk route connecting the Black Sea (Europe) to the Caspian (Central Asia). Latency to Frankfurt is excellent (~35-40ms), allowing for a hybrid Bitcoin + Edge Compute model.

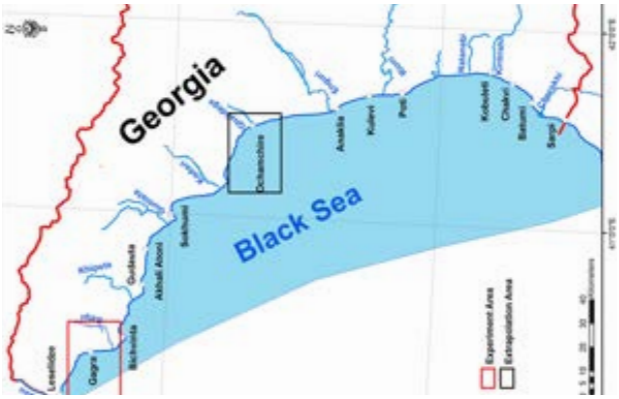
Seasonality Paradox

Winter Advantage: Excellent natural cooling (Free Cooling) from Oct–April reduces PUE to near 1.10.

Summer Trouble: Temp in Tbilisi/Kutaisi is 30°C+ (86°F+), needing cooling when power is affordable.

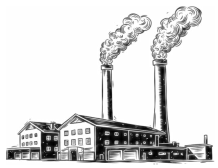
Georgia's grid is also the inverse of Ethiopia:

- **Enguri Dam (1,300 MW)** carries grid.
  - Summer (May-Sept): Water must be spilled; power is abundant/cheap.
  - Winter (Nov-Mar): Reservoir levels drop; Georgia imports gas/electricity
- **Winter Risk:** tariffs spike in 2024/25, the government has threatened to curtail miners first during deficits to protect residential heating.



MAP OF THE STRATEGIC LOCATION OF THE BLACK SEA IN CONTEXT TO THE MAJOR GEORGIAN CORRIDORS

COMPETITIVE ANALYSIS: DAVID VS. GOLIATH CASE IN INFRASTRUCTURE PENITRATION



BITFURY

The Incumbents:

Large conglomerate scaled miners

Example: Bitfury basically built the Georgian market.
They operate a 40MW+ immersion facility in Tbilisi FIZ (Free Zone)

"Home Miner" Crowd

Georgia has one of the highest per-capita crypto ownership rates. Thousands run rigs in basements.

Relevance: They create regulatory noise and public backlash against "energy waste."

Tatari's Positioning: "Seasonal Arbitrage & Tax Optimization." We do not enter Georgia for raw power (prices have risen to ~$0.05-0.06/kWh outside FIZ). We enter for regulatory certainty inside FIZs and to capture summer hydro dumping.

Thesis: Use Georgia as a tax HQ and summer-peak site, while maintaining Ethiopia for year-round base load.

THE MATURE MARKET PIVOT

SUMMARY

Georgia in December 2025 represents Tatari's flagship "Optimization" opportunity: Unlike Ethiopia's raw generation growth, Georgia offers a mature, tax-advantaged jurisdiction positioned as the digital gateway between Europe and Asia. However, it is a market defined by extreme seasonality—abundant hydro in summer, deficits in winter—requiring a fundamentally different operating model than Ethiopia's baseload GERD.

Georgia's Structural Realities:

1. **The "Free Zone" Advantage:** Operations within Free Industrial Zones (FIZ) (Tbilisi, Poti, Kutaisi) enjoy 0% Corporate Tax, 0% VAT, and 0% Dividend Tax. This creates a pure tax-arbitrage play unavailable in most jurisdictions.
2. **Hydro Seasonality Trap:** Unlike Ethiopia's GERD (steady baseload), Georgia's Enguri Dam relies on seasonal melt. The country has a Summer Surplus (May-Aug) but a severe Winter Deficit (Nov-Mar), often necessitating expensive gas imports or power rationing.
3. **The "Golden Era" Legacy:** Georgia was once the #2 miner globally (after China). The infrastructure (substations, fiber) exists, but the grid is now strained by "grey market" mining, particularly in the Abkhazia region, forcing the government to tighten regulations on non-FIZ operators.
4. **Geopolitical "Middle Corridor":** Georgia is positioning itself as the digital trunk route connecting the Black Sea (Europe) to the Caspian (Central Asia). Latency to Frankfurt is excellent (~35-40ms), allowing for a hybrid Bitcoin + Edge Compute model.

Tatari's Positioning:
"Seasonal Arbitrage & Tax Optimization." We do not enter Georgia for raw cheap power (prices have risen to ~$0.05-0.06/kWh outside FIZ). We enter for regulatory certainty inside FIZs and to capture summer hydro dumping.

Thesis: Use Georgia as a tax-efficient HQ and summer-peak mining site, while maintaining Ethiopia for year-round baseload.

I. PHYSICAL INFRASTRUCTURE ADVANTAGES:

A. Climate & Cooling: The Temperate Zone
While Ethiopia offers high-altitude "perpetual spring" for passive cooling, Georgia offers a temperate climate with distinct seasons.

- Winter Advantage: Excellent natural cooling (Free Cooling) from Oct–April reduces PUE to near 1.10.
- Summer Challenge: Temps in Tbilisi/Kutaisi reach 30°C+ (86°F+), requiring active cooling (evaporative/mechanical) exactly when power is cheapest.
- Tatari Competitive Edge: Deploy Immersion Cooling (already proven by Bitfury in Tbilisi) to mitigate summer heat and dust, ensuring 99% uptime during the profitable summer hydro season.

B. The Energy Grid: Feast or Famine
Georgia's grid is the inverse of Ethiopia's.

- The Enguri Factor: The Enguri Dam (1,300 MW) is the spine of the grid.
  - Summer (May-Sept): Water must be spilled; power is abundant and cheap.
  - Winter (Nov-Mar): Reservoir levels drop; Georgia imports gas/electricity from Azerbaijan/Russia to meet demand.

- The Winter Risk: Industrial tariffs spike in winter. In 2024/25, the government has threatened to curtail crypto miners first during winter to protect residential heating.

- Strategic Implication: Our financial model must assume curtailment or higher OPEX for 3 months/year. This differs from the 24/7 uptime in Addis Ababa.

COUNTRY DEEP DIVE #3

PARAGUAY

Tatari Systems

PARAGUAY

Paraguay in December 2025 represents a hydropower-abundant, policy-sensitive emerging compute market. The country combines some of the world's cheapest renewable baseload electricity with an underdeveloped but professionalizing digital infrastructure ecosystem.

Structural Advantages:

Hydroelectric dominance: Paraguay's electricity matrix is over 98% renewable, driven primarily by large-scale hydropower generation from Itaipú and Yacyretá. This provides continuous, non-intermittent base load electricity suitable for 24/7 compute operations.

Formal industrial tariff framework: The national electric utility, Administración Nacional de Electricidad (ANDE), has established a specific tariff category for large electricity consumers, including data centers and crypto-asset mining. This creates a legal, regulated pathway for scale.

Tri-Border geography: Paraguay's eastern industrial corridor near Ciudad del Este sits adjacent to Brazil and Argentina, enabling cross-border logistics, fiber routes, and a regional "export compute" strategy rather than reliance on domestic demand alone.

Early-stage but institutionalizing datacenter market: Paraguay has begun transitioning from ad-hoc server rooms toward certified colocation facilities, signaling new enterprise-grade infrastructure.

Key Constraints

- Regulatory scrutiny is increasing: Government attention is shifting toward monitoring and formalizing mining and large-load users, driven by concerns around illegal electricity consumption.
- Tariff volatility risk: Electricity pricing for intensive users has already been adjusted upward once, reinforcing the need for contractual protections.
- Challenging climate for cooling: Unlike Ethiopia's high-altitude advantage, Paraguay's hot and humid climate necessitates robust mechanical cooling, which increases capital and operating complexity.

Recent certification of a Tier III facility by Tigo Business (a subsidiary of Millicom) marks an inflection point toward professional colocate.

Tier III implies maintainability and redundancy that allow concurrent maintenance without a full shutdown.

The market is underbuilt, but it is no longer informal. This creates a narrow window for credible first movers to establish institutional trust.

Paraguay's electricity system is centrally managed by Administración Nacional de Electricidad (ANDE), the state-owned utility responsible for generation procurement, transmission, distribution, and tariff classification.

ANDE controls:

- Grid connections and substation access
- Tariff categories for residential, commercial, industrial, and special consumers
- Restrict electricity theft & unregistered loads

For Tatari, ANDE is the single most important counterparty. All power economics, curtailment risk, and regulatory exposure flow via ANDE's tariff classifications/ contracts.

In 2022, Paraguay enacted Decreto N° 7824/22, creating a special tariff category known as Grupo de Consumo Intensivo Especial (Special Intensive Consumption Group). This category includes:

- Data processing and data storage facilities
- High-density computing operations
- Cryptocurrency mining operations

Paraguay is best positioned as a hydro-powered export compute hub, not a speculative mining enclave. Tatari's value proposition is infrastructure-grade compliance, cooling discipline, and export-oriented compute services, with mining used tactically to monetize excess capacity.

THE SEASONALITY PARADOX

A. Climate & Cooling: The Temperate Zone

While Ethiopia offers high-altitude "perpetual spring" for passive cooling, Georgia offers a temperate climate with distinct seasons.

- **Winter Advantage:** Excellent natural cooling (Free Cooling) from Oct–April reduces PUE to near 1.10.
- **Summer Challenge**: Temps in Tbilisi/Kutaisi reach 30°C+ (86°F+), requiring active cooling (evaporative/mechanical) when power is cheapest.
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Georgia's grid is the inverse of Ethiopia's.

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  - Summer (May-Sept): Water must be spilled; power is abundant and cheap.
  - Winter (Nov-Mar): Reservoir levels drop; Georgia imports gas/electricity from Azerbaijan/Russia for heating demand.
- **The Winter Risk:** Industrial tariffs spike in winter. In 2024/25, the government has threatened to curtail crypto miners first during winter deficits to protect heating.
- **Strategic Implication:** Our financial model must assume curtailment or higher OPEX for 3 months/year.

THE "TWO" GEORGIAS

A. The Free Industrial Zones (FIZ) – The Only Viable Path

Georgia effectively operates as 2 different jurisdictions

- Mainland Georgia: Subject to 18% VAT and 15% Corporate Tax. Energy tariffs are standard rates. High Risk.
- The FIZ (Poti, Kutaisi, Tbilisi): The "Safe Harbor."
  - 0% VAT on electricity (lowering OPEX by 18%).
  - 0% Corporate Income Tax on international revenue (mining rewards are considered export services).
  - No Capital Controls: easy repatriation of USD/EUR.

B. The Abkhazia "Grey Market" Risk

- The Situation: The breakaway region of Abkhazia (occupied/autonomous) pays near-zero for electricity from Enguri Dam. This has led to massive illegal mining farms that destabilize the entire Georgian grid.
- The Risk: In winter, when Abkhazia drains the grid, the central government in Tbilisi is forced to cut power to legitimate industries to balance the load.
- Tatari Mitigation: We must locate in Kutaisi or Tbilisi FIZ (deep inside government-controlled territory) with dedicated substations, avoiding the unstable Western grid near the conflict line.

COMPETITION? DAVID VS. GOLIATH

A. The Incumbent: Bitfury

- Status: Bitfury built the Georgian market. They operate a 40MW+ immersion facility in Tbilisi FIZ. They have deep government ties and own their custom transformers.
- Implication: We cannot compete on "political access." We must compete on agility. Bitfury is legacy infrastructure. Tatari can deploy next-gen S21/S22 fleets more efficient per MW.

B. The "Home Miner" Crowd

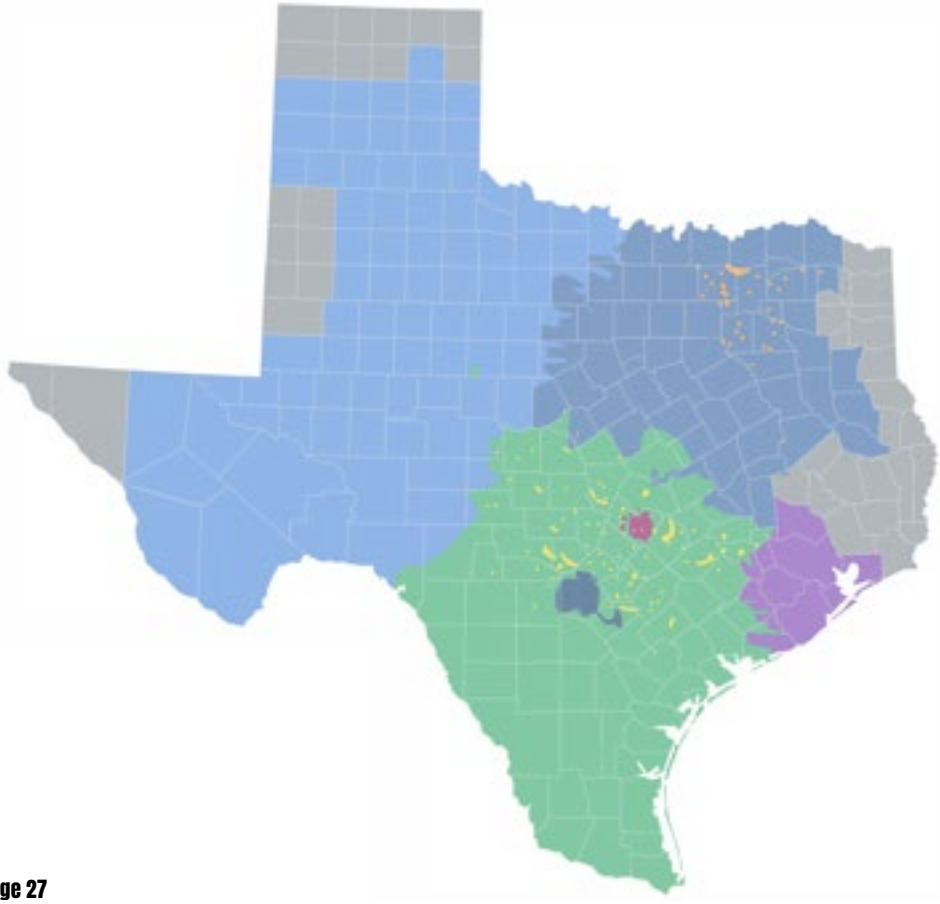
- Georgia has one of the highest per-capita crypto ownership rates. Thousands of home miners run rigs in basements.
- Relevance: They create regulatory noise and public backlash against "energy waste." Tatari must distance itself from residential mining and present as a "High-Tech Data Exporter" in the FIZ.

2025 EDITION

RESEARCH PAPER #1

Data Centers & Water Usage in Texas: Effects of New, Qualifying, Large Data Centers on County-Level Water Withdrawals & Groundwater Reliance

Amen Amare
Jaemoon Lee



TATARI SYSTEMS

Abstract

Data centers now play a key role in digital systems; however, their reliance on water for cooling brings issues where fresh water is getting scarcer. This study explores if growing numbers of big, tax-advantaged data hubs in Texas led to measurable shifts in local water use across counties. Drawing on a detailed dataset covering all 254 counties between 2000 and 2023, we merge official reports on water consumption, measurements of underground pumping, population trends, along with state records of eligible sites. By leveraging the uneven rollout of certifications under Tax Code §151.359, we apply the Callaway–Sant'Anna method to assess impacts over time. In terms of total withdrawals, groundwater extraction, or urban supply, post-policy impacts appear minimal - statistically near zero - with event analyses revealing stable trends before implementation and no lasting changes afterward. This implies certified data hubs so far haven't significantly affected local water availability across Texas counties.

1. INTRODUCTION

Large data centers have emerged as essential infrastructure for digital services and machine learning, but they also concentrate significant power and cooling demands in a limited number of locations. Many of these facilities rely on water-based cooling systems, resulting in substantial annual water withdrawals–often in areas where freshwater resources are projected to decline. At the same time, projections indicate that groundwater availability in Texas could decrease by approximately 25 percent and total water supplies by about 20 percent by 2070.

This study investigates whether tax-favored data centers affect county-level water use in Texas. An annual panel dataset is constructed for all 254 counties from 2000 to 2023, integrating withdrawal records from the Texas Water Development Board–including total withdrawals, groundwater pumpage, and municipal use–with demographic and economic indicators from the Bureau of Economic Analysis.

Recent plans show widening gaps between projected demands and available surface and groundwater in several regions, even before accounting for new, potentially water-intensive sectors. At the same time, Texas has long used targeted tax incentives to attract large investments. In 2013, lawmakers introduced a sales-tax exemption for "Qualifying Data Centers" under Tax Code §151.359, granting exemptions on specified purchases if facilities meet thresholds for floor space and capital expenditure. The Comptroller maintains a registry of qualifying projects, recording the locations and effective dates of certification; that registry provides the policy variation analyzed in this study.

To examine the relationship between these facilities and water use, a county–year dataset covering 2000–2023 is constructed. For each county and year, sector-level withdrawal records from the Texas Water Development Board are aggregated into three outcomes, measured in acre-feet: total withdrawals, groundwater pumpage, and municipal withdrawals.

The Comptroller registry is merged to identify, for each county, the year of its first qualifying data center and the annual number of qualifying facilities in operation. The arrival of the first qualifying data center is treated as a staggered intervention, with counties receiving a qualifying facility between 2013 and 2023 compared to those whose first facility appears only in 2024 or 2025. The latter group remains untreated during the 2000–2023 analysis window and serves as future-treated controls.

To estimate causal effects, the Callaway and Sant'Anna (2021) difference-in-differences estimator is applied, which calculates group-time average treatment effects by comparing each treated cohort to not-yet-treated counties in each year, and then aggregates these into overall post-treatment effects and event-time coefficients. The primary identifying assumption is that, in the absence of treatment, treated and future-treated counties would have exhibited similar trends in water outcomes.

The primary finding is that qualifying data centers are associated with small and statistically imprecise changes in county-level water use. In logarithmic terms, the estimated average treatment effect for total withdrawals is approximately −0.03 log points (about a 3 percent decline), while groundwater pumpage increases by about +0.048 log points (around 5 percent) and municipal withdrawals by about +0.013 log points (approximately 1 percent). None of these estimates is statistically distinguishable from zero at conventional significance levels. Considering that typical annual withdrawals in many Texas counties are in the tens of thousands of acre-feet, these bounds indicate that very large average effects of qualifying data centers on aggregate demand can be ruled out. Additionally, another key result is that dynamic treatment effects do not indicate strong pre-trends or persistent post-treatment shifts. After treatment hits, numbers don't climb or drop consistently; instead, they bounce around baseline levels, hinting that adding a qualified data center doesn't shift how county withdrawals evolve during this time frame. For groundwater use and city-level draws, you see roughly the same behavior - but error bars stretch further out here.

2. DATA

Data Sources and Panel Construction

We compiled and aggregated a total of 5 datasets to make a county-year panel of all 254 counties in Texas, 2000-2023. The former is the Texas Water Development Board (TWDB) Historical Water Use Estimates that lists annual county-level water withdrawals at the sector level. These files give the municipal, manufacturing, mining, power, irrigation, and livestock volumes in acre-feet. The result of this process is a constant county-sector-year panel which we collapse to county-year totals.

There is another TWDB dataset of groundwater pumpage at reported county-aquifer-sector level initially. We annualized imports, eliminated non-numeric data and totaled across aquifers to provide county totals of manufacturing, irrigation and municipal groundwater pumpage. These were then combined with the withdrawals panel with county and year attributes.

Variable	n	Mean	Std. Dev.	25th pct	Median 50th	75th pct
Total withdrawals	720	138,101	214,546	23,578	58,832	145,412
Groundwater pumpage	720	42,857	70,390	7,728	14,665	45,083
Municipal withdrawals	720	98,584	162,834	4,709	25,265	117,496

TABLE 1: OVERALL LEVELS (LEVELS, ACRE-FEET)

Table 1 summarizes the distribution of county-year withdrawals for our three main outcomes. Average total withdrawals are about 138,000 acre-feet per county-year, but the standard deviation is more than 200,000 acre-feet and the interquartile range runs from roughly 24,000 to 145,000 acre-feet. Groundwater pumpage and municipal withdrawals display similarly wide dispersion. Median values sit well below the means in every case, which is consistent with a right-skewed distribution where a small number of large, water-intensive counties drive much of the aggregate. This variation in scale is one reason we work primarily with logged outcomes in the empirical analysis.

The data of population and income are provided by the Bureau of Economic Analysis (BEA) CAINC1 Local Area Personal Income series. Then we control the log population and the log income in the empirical analysis. The final input is the registry of Qualifying Data Centers (QDC) and Qualifying Large Data Center Projects (QLDC) of the Comptroller. Under Tax Code SS151.359, the registry indicates county, operator & certification date.

Following the cleaning of the county names and the combination of the FIPS codes to the county level, we discover (i) year of the first qualifying data centre and (ii) the number of qualifying data centres in operation in each county-year, taking into account those effectively in operation during their year.

Variable Construction

The main water demand measure that will be utilized in the analysis is the total annual withdrawals, labeled as wu total,af, which is defined as the total of municipal, manufacturing, mining, power, irrigation, and livestock withdrawals as given by the Texas Water Development Board (TWDB). Since these sectoral volumes are given in different files and cover several years, we converted all the quantities to acre-feet, standardized county identifiers and condensed the data to county level.

The use of groundwater is measured in units of gw, pump total af, which is the sum of the pumpage of all the aquifers of each county-year combination. The raw pumpage data is first stored at the county-aquifer-sector granularity; thus, aggregated into county year values following reprocessing of formatting anomalies and equalization of numeric data. Municipal withdrawals, denoted as muni af, are derived directly out of municipal sector totals in the Texas Water Development Board (TWDB) water use files and form an outcome measure potentially sensitive to large industrial customers, such as data centers.

Descriptive statistics

We start by looking at treated versus future-treated areas before any data center shows up. In every county-year combo from the Callaway–Sant'Anna dataset, nearly all, 95.7%, saw no active facilities. Just a small slice, about 4.3%, included at least one.

Variable	Not Yet Treated (0)	Early Treated (1)	Overall Mean
Total withdrawals	93,612	162,426	137,194
Groundwater pumpage	29,043	49,323	41,887
Municipal withdrawals	58,101	119,880	97,228

TABLE 2: EARLY VS. FUTURE-TREATED MEANS

Table 2 compares mean withdrawals in early-treated and future-treated counties before any qualifying data center appears. Early-treated counties have higher pre-treatment totals: average total withdrawals are about 162,000 acre-feet versus 94,000 in future-treated counties, with similar gaps for groundwater and municipal use. At the same time, the overall mean lies in between the two groups, and the differences are small relative to the very wide dispersion in Table 1.

We interpret these patterns as showing that early-treated counties are somewhat larger and more urban on average but still comparable to future-treated counties along the main water outcomes. Together with the narrow differences in Table 1, this supports using future-treated counties as a control group in the Callaway–Sant'Anna design.

3. EMPIRICAL STRATEGY

Treatment Timing and Comparison Groups

The empirical strategy we use takes advantage of the fact that qualifying data centers will be rolled out at a staggered pace, one county at a time. We define each county's treatment cohort by the year in which it first becomes eligible for the tax exemption under Texas Tax Code §151.359.

Let Gc denote the first qualifying data center year for county c. This variable divides counties into cohorts that adopt treatment at different times, mirroring a staggered rollout. We classify early-treated counties as those with:

$$G_c \in [2013, 2023]$$

This way, the comparison for each treated cohort has counties that receive qualifying data centers at some point but haven't yet reached their own first treatment year.

Estimator and Identification

A central challenge in studying the effects of qualifying data centers is that counties adopt treatment at different times. Under staggered adoption, a standard two-way fixed effects (TWFE) difference-in-differences regression can produce biased and misleading estimates. TWFE implicitly pools all treated observations together, regardless of when treatment occurs, and often uses already-treated units as controls for later-treated units. When treatment effects vary over time or differ across cohorts, the resulting estimates become weighted averages of incompatible comparisons, potentially even assigning negative weights to certain groups.

In order to address these issues, the Callaway Sant Anna (2021) estimator will be used, which is specially designed to support staggered intervention schedules.

Let Yct denote a water outcome (e.g., log total withdrawals) in county c and year t, and let Gc be the first year county c receives a qualifying data center. Following Callaway and Sant'Anna (2021), we index treatment groups by their first treatment year g and define potential outcomes under different treatment timings.

Then the group-time ATT:

$$ATT(g,t) = E\left[Y_t^g - Y_t^\infty \mid G_c = g,\ t \geq g\right],$$

Where Yt g is the potential outcome in year t if a county is first treated in year g, and Yt is the potential outcome if it is never treated. In practice, we approximate the never-treated counterfactual using future-treated counties that have not yet received a qualifying data center by year t. The estimator does not estimate a homogeneous pooled effect, but rather, it results in group-time specific average treatment effects (ATT(g,t)) the effect on the cohort first treated in year g, measured in period t, compared to counties not yet treated by that time. After the set of ATT(g,t) estimates is available, the process synthesises them by use of two different aggregation schemes:

1. **Overall post-treatment ATT**, that spreads ATT(g, t) over groups and later treatment phases with suitable scaling - also blending in proper adjustments
2. **Event-study coefficients**, that adjusts timing depending on when each county received treatment, revealing effects before plus after a qualified data center launched.

This estimator answers if:

For counties that first get a qualifying data center in year g, how do water outcomes change before and after treatment versus counties that receive centers later but not yet?

Estimation Details

For intuition, it is useful to write a standard two-way fixed-effects specification that nests the components used by the Callaway–Sant'Anna estimator:

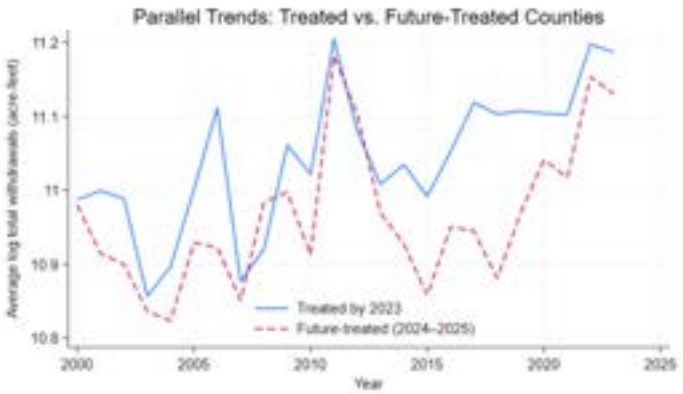
$$Y_{ct} = \alpha_c + \gamma_t + \beta_{g(t)} D_{ct} + X_{ct}'\delta + \varepsilon_{ct},$$

where c are county fixed effects, t are year fixed effects, Dct is an indicator that county c has at least one qualifying data center in year t, and g(t) captures the effect for the cohort first treated in year g at calendar time t. The vector Xct includes log population and log per-capita personal income, and ct is an idiosyncratic error term.

In practice, the Callaway–Sant'Anna estimator does not estimate a single pooled g(t); instead, it recovers the group–time effects ATTg,t in equation (1) by comparing treated counties to not-yet-treated counties within each year, and then aggregates them into overall and event-time estimates.

County fixed effects c absorb time-invariant characteristics such as terrain, long-run water availability, and baseline industrial structure. Year effects t capture statewide shocks like drought episodes, regulatory changes, and aggregate economic conditions. Clustered standard errors at the county level account for arbitrary serial correlation in ct within counties over time. We use log values for county population and income per person as control factors. Since these reflect broader social and economic trends, they might affect water use on their own.

Assumptions and Limitations



Our strategy relies heavily on the parallel trends assumption, as we count on the idea that untreated areas would've moved similarly, early ones versus later ones, when it comes to results. Figure 1 plots average log total withdrawals from 2000 to 2023 for counties treated by 2023 and counties whose first qualifying data center appears only in 2024–2025. The two series move closely together over the full window, with no systematic divergence in the years leading up to treatment.

Both groups exhibit similar year-to-year fluctuations, which largely reflect statewide shocks such as drought episodes and business cycles. This visual evidence is consistent with the idea that, absent treatment, early- and future-treated counties would have followed comparable trends in withdrawals, which is the core parallel-trends condition underlying our research design.

4. KEY RESULTS

Parallel Trends and Event-Time Patterns

We assess the identifying assumption of parallel trends using the event-study estimates from the Callaway–Sant'Anna estimator. In all specifications, pre-treatment coefficients lie close to zero and fall well within the 95% confidence intervals, consistent with the assumption that early- and future-treated counties followed similar trends before treatment. We see this in both the raw pre-period averages as well as event-time coefficients.

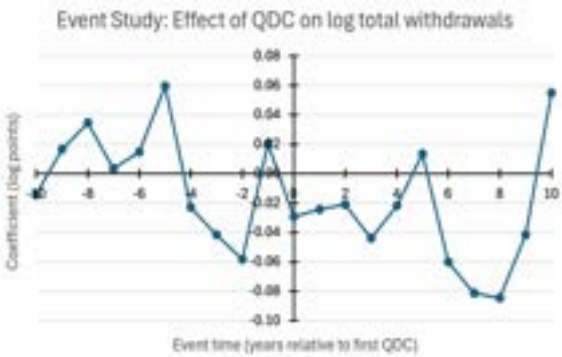


FIGURE 2: EVENT-STUDY ESTIMATES FOR LOG TOTAL WITHDRAWALS

Figure 2 reports event-time coefficients from the Callaway–Sant'Anna estimator for log total withdrawals, re-indexed so that event time 0 is the year of a county's first qualifying data center. The pre-treatment coefficients for event times −10 through −1 cluster near zero and do not display any clear upward or downward trend, which provides additional support for the parallel-trends assumption. Post-treatment coefficients for event times 0 through +10 fluctuate around zero without a persistent shift in either direction, and their magnitudes are small, mostly within roughly five percentage points. Combined, the dynamic pattern suggests that the arrival of a qualifying data center has not yet produced sustained changes in county-level withdrawals over the decade following treatment.

Outcome	ATT in logs (log points)	p-value: P>|z|	Approx. % Change	ATT in levels (acre-feet)	95% CI (levels)
Total Withdrawals	-0.0307	0.335	≈ −3%	+812.06	[−9,518.29 , 11,142.41]
Groundwater Pumpage	0.0483	0.456	≈ +5%	+882.73	[−5,213.28 , 6,978.74]
Municipal Withdrawals	0.0129	0.695	≈ +1%	+2,926.55	[−5,832.93 , 11,686.03]

TABLE 3: MAIN CALLAWAY–SANT'ANNA RESULTS (LOGS & LEVELS)

Table 3 reports average post-treatment effects from the Callaway–Sant'Anna estimator. Across all three outcomes: total withdrawals, groundwater pumpage, and municipal withdrawals, we see no statistically significant evidence of change after the arrival of a qualifying data center:

- **Total withdrawals (log):** ATT ≈ −0.031 log points (≈ −3%), p ≈ 0.335
- **Groundwater pumpage (log):** ATT ≈ +0.048 log points (≈ +5%), p ≈ 0.456
- **Municipal withdrawals (log):** ATT ≈ +0.0129 log points (≈ +1%), p ≈ 0.695

In level terms, the corresponding confidence intervals for total withdrawals span from −9,518 to +11,142 acre-feet per county-year, with similarly wide but centered intervals for groundwater and municipal use. These estimates are small in magnitude and statistically indistinguishable from zero, indicating no detectable average shift in county-level withdrawals after a qualifying data center is certified.

Robustness: Intensity & QLDC-Treatment Specifications, and TWFE Comparison

Panel A: Intensity Model (DC Count Control)

```
. di "Intensity model: ln_wu_total_af with n_dc_t control"
Intensity model: ln_wu_total_af with n_dc_t control

. estat simple
Average Treatment Effect on Treated
```

	Coefficient	Std. err.	z	P>|z|	[95% conf.	Interval]
ATT	-.0307434	.0318858	-0.96	0.335	-.0932385	.0317517

Panel B: QLDC-Only Treatment

```
. di "QLDC-only ATT: ln_wu_total_af"
QLDC-only ATT: ln_wu_total_af

. estat simple
Average Treatment Effect on Treated
```

	Coefficient	Std. err.	z	P>|z|	[95% conf.	Interval]
ATT	-.0416355	.0497985	-0.84	0.403	-.1392387	.0559678

Panel C: TWFE Comparison

Outcome (log)	Coefficient on dc_any	Robust SE	p-value
Total Withdrawals	−0.0414	0.0691	0.554
Groundwater Pumpage	−0.0073	0.0917	0.937
Municipal Withdrawals	0.0399	0.0484	0.417

As a first robustness check, Panel A reports an intensity specification in which the Callaway–Sant'Anna estimator controls for the number of active qualifying data centers rather than just the first arrival.

The ATT on log total withdrawals in this model is −0.0307 with a standard error of 0.0319 ($p = 0.335$), nearly identical to the baseline CS estimate. The 95-percent confidence interval ranges from about −0.093 to +0.032 log points. This suggests that allowing the treatment effect to vary with the count of operating centers does not materially change the conclusion that qualifying facilities have, at most, small average effects on county-level withdrawals.

Panel B restricts the analysis to Qualifying Large Data Center Projects (QLDCs), which satisfy higher thresholds for investment and floor space. In this subsample the ATT for log total withdrawals is −0.0416 with a standard error of 0.0498 ($p = 0.403$), implying an imprecisely estimated 4 percent decline. The point estimate is somewhat more negative than in the baseline specification, but the confidence interval comfortably includes zero and remains far from any large decline. Given the small number of QLDC counties, these results reinforce the view that even the largest tax-favored data centers have not produced substantial changes in aggregate county-level water use.

Panel C reports simple two-way fixed-effects regressions of each log outcome on an indicator for any qualifying data center, controlling for county and year fixed effects and the same time-varying covariates. The coefficient on dc_any is −0.0414 for total withdrawals (standard error 0.0691; $p = 0.554$), −0.0073 for groundwater pumpage (standard error 0.0917; $p = 0.937$), and 0.0399 for municipal withdrawals (standard error 0.0484; $p = 0.417$). As with the Callaway–Sant'Anna estimates, these effects are small in magnitude and statistically insignificant. The TWFE model therefore leads to the same substantive conclusion of no meaningful average effect, but it masks potential cohort- and time-specific variation and, as recent work shows, can assign problematic weights when treatment is staggered.

Presenting both sets of estimates illustrates that our main findings are not an artefact of the newer estimator, while also underscoring why we rely on Callaway–Sant'Anna for interpretation.

5. DISCUSSIONS OF RESULTS, CONTRIBUTIONS, & RECOMMENDATIONS

Interpretation & Limitations

A limitation to note is grouping by county and year hides differences in water systems and pipes. Instead of showing real impacts, it smooths out problems, like when a big server farm taps into a single city's supply, well zone, or underground layer, pushing stress locally while overall county use stays flat. That matters most in big counties mixing cities with open land. Since our setup doesn't separate usage from strained groundwater versus rivers, it misses spikes or time-based strains that affect service.

In addition, we look only at sites eligible for the §151.359 program. Tiny data centers, edge locations, or crypto mining spots under the size or spending limits aren't listed, still they might use local water. When those left-out places cluster near approved ones, our measure misses some digital load in affected areas. In the end, just like any diff-in-diff setup, this method assumes early and late participants would've moved alike without intervention, as long as fixed traits and variables stay accounted for. Even though steady pretrend patterns help ease doubts, hidden planning moves tied to new water shifts could still skew results.

Continuations and Policy Relevance

Even with these limits, what we found matters for how Texas handles water and infrastructure. The lack of strong effects means the Qualifying Data Center program hasn't pushed counties into worse water strain overall so far. That lines up with forecasts from the 2022 State Water Plan, where experts say rising needs come mostly from more people, farming, and power plants, not data centers. Still, supplies are expected to shrink in the years ahead, making every drop count. Meanwhile, Texas keeps pulling in new data centers using tax breaks. The steady climb in water pressure along with faster growth of data centers calls for early action even when today's effects seem small based on available numbers. A clear takeaway? Water factors need a bigger role in how tax perks are shaped or where buildings go up. As the state accountant already showed, some areas may run short well before 2070; matching tech-building rewards with these realities can help stop fresh factory demands from making shortages worse.

This shows we really need more solid info. What we've seen is just how scarce details are on how much water individual plants use in data centers even though people keep digging into their power usage (Shebbie et al. 2016). If places shared yearly and max water draw regularly by site or service zone and included notes on cooling types and supply origins like rivers, wells, or recycled flows it'd give analysts and city planners a clearer edge over broad county stats. Tying this info into TWDB's modeling plus the State Plan could push RWPGs to weigh different setups for packs of AI-heavy server farms in certain watersheds.

Future research might go a few ways. A logical next step involves combining official water records with close-up looks at particular server farm centers, checking how power providers paid for added capacity, if they moved to recycled sources or saltwater processing, while also seeing what impact these moves had on nearby users. Alternatively, comparing Texas to areas using alternative rules like strict limits on underground withdrawals or transparency mandates for tech facilities' consumption can show how laws influence where companies build and what systems they pick. As both approved and unapproved computing sites expand in size and count, updating local assessments becomes key, bringing in broader groups exposed to changes, deeper metrics of usage rates, plus results tied clearly to shortages, say aquifer depths, reservoir heights, or enforced cutbacks. Our results aren't saying data centers don't matter when it comes to water rules. They just show that right now, the impact from programs like this one's pretty minor compared to bigger trends on water use.

Conclusion

Our study investigates if big data centers with tax advantages change how much water counties use across Texas. We look at yearly county data from 2000 to 2023, matching records on water draw and pumping with official lists of approved data centers. We apply a flexible diff-in-diff approach designed for staggered rollouts. Results show no clear boost in overall water use after a major facility opens. Groundwater extraction stays steady; city usage doesn't shift either. Our margins exclude swings in regional demand.

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ECONOMICS OF COMPUTE IN FRONTIER MARKETS

A major hidden constraint is currency and capital controls, which can trap cash, raise imported-equipment costs, and force operational and contracting changes, as seen in places like Ethiopia's cheap hydro but strict forex rules versus Nigeria's fuel-linked volatility and the UAE's capital-and-solar advantage. Tatari's angle is to treat compute like an industrial asset, pricing decisions on total cost per hour (including depreciation and uptime) and designing contracts to match currencies or index terms so returns aren't quietly eroded by FX and system fragility.

Tatari Systems & Castlecliffe Consulting: Alma de La Tournelle, Edison Zhou, George Yu and Jamie Campbell-Mooney

HARDWARE SCARCITY:

Introduction:
Artificial Intelligence. Both terrifying and vastly intricate. While AI was conceptually defined at The Dartmouth Conference of 1956, the true catalyst for its exponential growth was GPU-accelerated deep learning. Whilst software breakthroughs such as back propagation and Transformer architecture unlocked new capabilities, it was the rapid scaling of GPU compute that allowed AI models to grow by orders of magnitude. In order to achieve success, it is has become evidently vital to understand the environment of today's hardware scarcity.

Identifying Key Foundries:
The AI hardware supply chain is mainly dominated by a small number of advanced foundries. As we know, TSMC and, to a lesser extent, Samsung Foundry are manufacturing almost all AI accelerators for NVIDIA, AMD and other vendors. TSMC alone accounts for roughly one-third to forty percent of global foundry revenue and an even larger share of advanced nodes used for AI. Its latest disclosures indicate that advanced nodes and advanced packaging technologies are running at high utilisation as AI demand accelerates (Carbon Credits). This concentration means that any capacity constraint or disruption at TSMC can directly affect the availability and pricing of advanced AI and high-performance computing hardware.

Bottlenecks & Context of Market:

The current consequence for AI compute buyers such as Tatari systems are long and volatile lead times on data-centre GPUs. From late 2023 through mid-2024, H100 lead times for many enterprise customers extended to six to eleven months (order to delivery), reflecting both water constraints and limited CoWoS packaging capacity (Tom'sHardware).

While supply has eased, typical delivery windows for H100 systems in 2025 are still quoted in the range of 8–16 weeks, and some solution providers report system lead times "exceeding 40 weeks" depending on configuration (Jeskell Systems).

AMD's Instinct MI300 family is going through a similar dynamic. AMD has repeatedly raised its 2024 revenue guidance for MI300 to over $4–$4.5 billion, signaling very strong demand. Much industry analysis notes that AMD's lead times will remain extended as it ramps production into a supply-constrained market (The Next Platform). All of this implies that capacity expansions based on top-tier GPUs will almost always involve a multi-quarter planning horizon, which is definitely not just a purchasing decision.

Hyperscaler Favouritism and Hierarchies:
Global AI compute is structured around a clear hierarchy. At the top sit the hyperscalers, including AWS, Microsoft Azure, and Google Cloud. They control the majority of cloud IaaS revenue and are the primary counterparties for NVIDIA, AMD, and leading foundries. Recent market estimates show that the IaaS market is expanding rapidly (over 20% year-on-year), with most of this growth captured by these three firms and with a smaller residual segment available to alternative providers (Cloud IaaS Market Grows 22%).

Hyperscalers typically secure long-term capacity agreements directly with chip vendors and foundries, sometimes complemented by in-house accelerators (TPUs at Google, AWS chips, Maia/Cobalt at Microsoft), which further reinforce their bargaining power and priority access to advanced nodes (Engineering at Meta). Below the hyperscalers is a fast-growing layer of specialised GPU cloud providers such as CoreWeave, Lambda, and RunPod.

These firms focus almost exclusively on AI-ready GPU compute and have built tightly integrated clusters of H100/A100-class hardware with high-speed networking, which are aimed at AI labs and model companies (Top 12 Cloud GPU Providers for AI and ML 2025). NVIDIA has, of course, deepened this tier by investing in and partnering with several of these players, organising large GPU leasing and resale arrangements such as Lambda's recent 18,000-GPU leaseback deal —effectively using them as distribution and utilisation channels for its own silicon (Nvidia signs $1.5 billion deal with cloud startup Lambda to rent back its own AI chips). This creates a "second ring" around the hyperscalers who can still access high-end GPUs in bulk, but have less integrated ecosystems and weaker balance sheets.

100,000 Nvidia H100s

Everyone else—including regional clouds, telcos, colocation providers, and independent GPU-as-a-Service (GPUaaS) operators—typically operates downstream of these two tiers. They face a trickle-down dynamic as chip allocations and delivery schedules are set after serving hyper scalers and preferred partners, translating into higher effective prices, longer and/or less predictable lead times, and limited ability to negotiate custom SKUs and long-term supply guarantees (Breaking the GPU stronghold: emerging competition in AI infrastructure).

At the same time, demand is becoming increasingly concentrated as major foundation-model providers are now signing multi-billion-dollar cloud and capacity commitments such as OpenAI's multi-year AWS deal, which further entrenches hyperscalers as the default home for large-scale training workloads (How OpenAI's $38 Billion AWS Deal Reshapes AI Infrastructure and Market Dynamics). For an Ethiopia-based operator, these dynamics imply three structural realities:

1. Irregular access to cutting-edge GPUs as they are priority-biased, especially during global demand spikes.

2. Price formation is set upstream, where operators outside the hyper scaler ecosystem can expect periodic supply shocks.

3. Competitive advantage cannot be built on direct parity with hyper scalers, but instead operators differentiate through cost structure, geography, regulatory positioning, and workload focus.

ENERGY, INFRA, & CURRENCY AS COST DRIVERS

What limits growth in computing today isn't chips - it's how much it costs to run them. Power needs, heat management, and where money flows matter more now than ever. With heavy-duty computing tasks on the rise, certain emerging regions start making sense for big installations - assuming their grids hold up, weather helps, and financial climates stay steady. Instead, it's viewed like a factory process: expensive to build, shaped heavily by electricity prices, cooling effectiveness, and whether local currencies can reliably shift. What drives computing costs in emerging markets isn't just hardware or power bills - it's where investment outlays meet real-world logistics and money flows. Electricity rates might grab attention first, yet sustained performance ties more closely to actual energy expenses after transmission losses, how well systems shed heat, consistent uptime, and whether returns stay stable despite shifting exchange rates.

Capex and Opex in Frontier Contexts

Compute infrastructure exhibits a high fixed-cost profile globally, but frontier markets alter the relative weight and risk of those costs. Capital expenditures include electrical infrastructure, site development, cooling systems, IT hardware, and resilience equipment such as backup generation and power conditioning. In developed markets, these assets are largely standardized and benefit from mature supply chains and predictable grid performance. In frontier markets, however, CAPEX is more sensitive to site-specific conditions, including grid interconnection requirements, local construction capacity, customs regimes, and security considerations. Fuel bills shape most of the ongoing expenses, standing out across a system's lifespan. Still, in emerging regions, daily operations carry extra burdens - unstable supply needs careful handling, blackouts demand workarounds, equipment repairs depend on patchy delivery routes, rules shift under changing laws. Notably, loans tied to setup spending tend to cost more there, driven up by country-specific risks and scarce options for stable, long-term funding, making it harder to dodge repeated charges.

What makes Tatari's method stand out is how it treats computing expenses - not just as initial investments, but as ongoing commitments measured in total cost per hour. By folding both hardware depreciation and daily operations into one metric, decisions shift away from short-term discounts. Hidden upkeep demands won't get brushed aside so easily when they're baked into the core number from the start.

Location Arbitrage Beyond Headline Power Prices

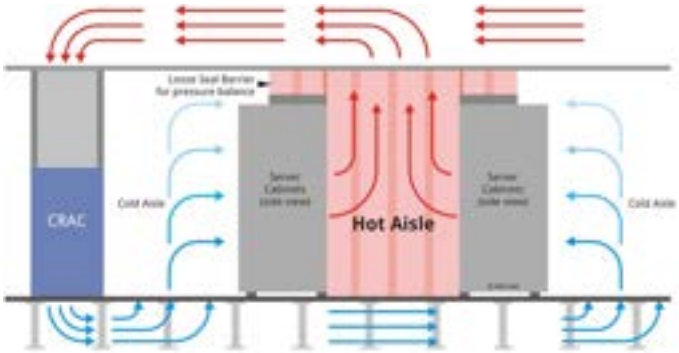
What makes computing viable in emerging economies isn't just cheap power - it's deeper structural gaps across infrastructure systems. Instead of mere price differences in electricity, it's the full ecosystem of supply that matters. Power delivered depends on more than meter rates; losses during transit play a role. Charges tied to peak usage add up. When service falters, extra costs emerge - like maintaining standby generators. In places with shaky grids, blackouts turn inexpensive rates into something far pricier. Depending on fuel-powered backups quickly changes the math, even if base prices look favorable.

Where things are built affects how well they stay cool. Most of the power used by data centers turns straight into heat - so handling that heat shapes much of what it costs to run them. Cooler climates, along with balanced moisture levels in the air, make it possible to pull in outside air instead of relying on heavy-duty chillers. That shift cuts down on mechanical cooling, which means less energy gets burned overall. Over months or years, these conditions slowly add up: systems run more efficiently, hardware lasts longer, and each kilowatt-hour does more work than before.

What often gets overlooked is how money movement rules twist the actual gains from what looks like a pricing gap. A business might earn dollars on paper, yet find itself stuck - unable to move profits out or swap currencies, which eats into margins, messes with deliveries, limits growth steps. Because of this, calculating true costs in emerging tech markets means seeing hardware, red tape, and cash flows as tangled pieces, not just scanning sticker prices.

Cooling Dynamics and the Role of Altitude

Thermal performance is often underestimated in early-stage site selection, yet it is one of the most persistent contributors to cost differentials across markets. In higher-altitude regions, such as the Ethiopian highlands, average temperatures are materially lower than in many equatorial or desert environments, increasing the number of hours per year during which free cooling is viable.



While reduced air density at altitude can necessitate higher airflow volumes to achieve equivalent heat transfer, this effect is frequently outweighed by the benefits of cooler intake air and lower compressor runtimes. For high-density compute, including AI and high-performance workloads, liquid cooling architectures further mitigate altitude-related airflow constraints by decoupling heat removal from ambient air properties. In such configurations, climate becomes an advantage rather than a limitation, particularly when paired with low-cost, renewable energy.

Currency Risk & Serialization of FX Constraints

Perhaps the most under appreciated risk in frontier compute economics is the way foreign exchange constraints propagate through operating models. Currency risk does not appear only at the point of conversion; it serializes across tariffs, supply chains, and capital allocation decisions. Utilities facing dollar-denominated debt or imported equipment costs may re-index tariffs, increasing local currency charges. Spare parts, networking equipment, and cooling components are typically imported, requiring access to hard currency regardless of where revenue is generated. Cash stuck in local markets - often due to capital limits or controlled currency systems - can slow down payments abroad or hinder expansion plans. Take Ethiopia: known for strict forex rules, it pushes companies to rethink how they draft agreements, manage liquidity, or handle cross-border payouts. When exchange hurdles aren't planned for, even promising energy ventures may struggle just keeping up with daily demands.

What sets Tatari apart? Matching currencies whenever feasible - this isn't a blanket fix, though. When that doesn’t work, contracts shift toward indexed terms. Flexibility in operations helps maintain function and protect assets even when economic conditions wobble. The goal here isn't some illusion of total safety. It's clearer pricing of what risk remains. Returns stay intact, regardless of which currency landscape shifts beneath them.

Conclusion: System-Level View of Compute

A fresh look at computing economics in emerging regions shows it's never just about one factor. Instead, power expenses, weather patterns, network stability, and financial policies mix - shaping what lasts over time.

Take Ethiopia: its reliance on hydropower means electricity stays cheap at the margin, especially if systems use natural cooling smartly. In contrast, Nigeria leans on gas, revealing how swings in fuel supply and weak grids create hurdles.

Then there’s the UAE, where big investments and solar capacity drive down energy costs dramatically - even as they demand sharper engineering for heat control and funding structures. What sets Tatari Systems apart begins with how it treats computing power - not as some invisible cloud resource, but like steel mills or rail lines of the digital age.

Seeing computation through an industrial lens shifts everything. Where others spot data centers, Tatari sees leverage points - places where land access, cooling methods, and local electricity rates combine unexpectedly. These aren't random wins; they're patterns shaped by geography meeting capital design. In emerging tech regions, that mix can lock in lower operating costs for years. Physical edges turn into economic ones when handled right. Infrastructure, priced smart today, becomes hard to displace tomorrow.

2025 EDITION

RESEARCH PAPER #2

Applications of Stablecoins in Emerging Markets: Reviewed and Edited by the Economics Department of Boston College

Amen Amare
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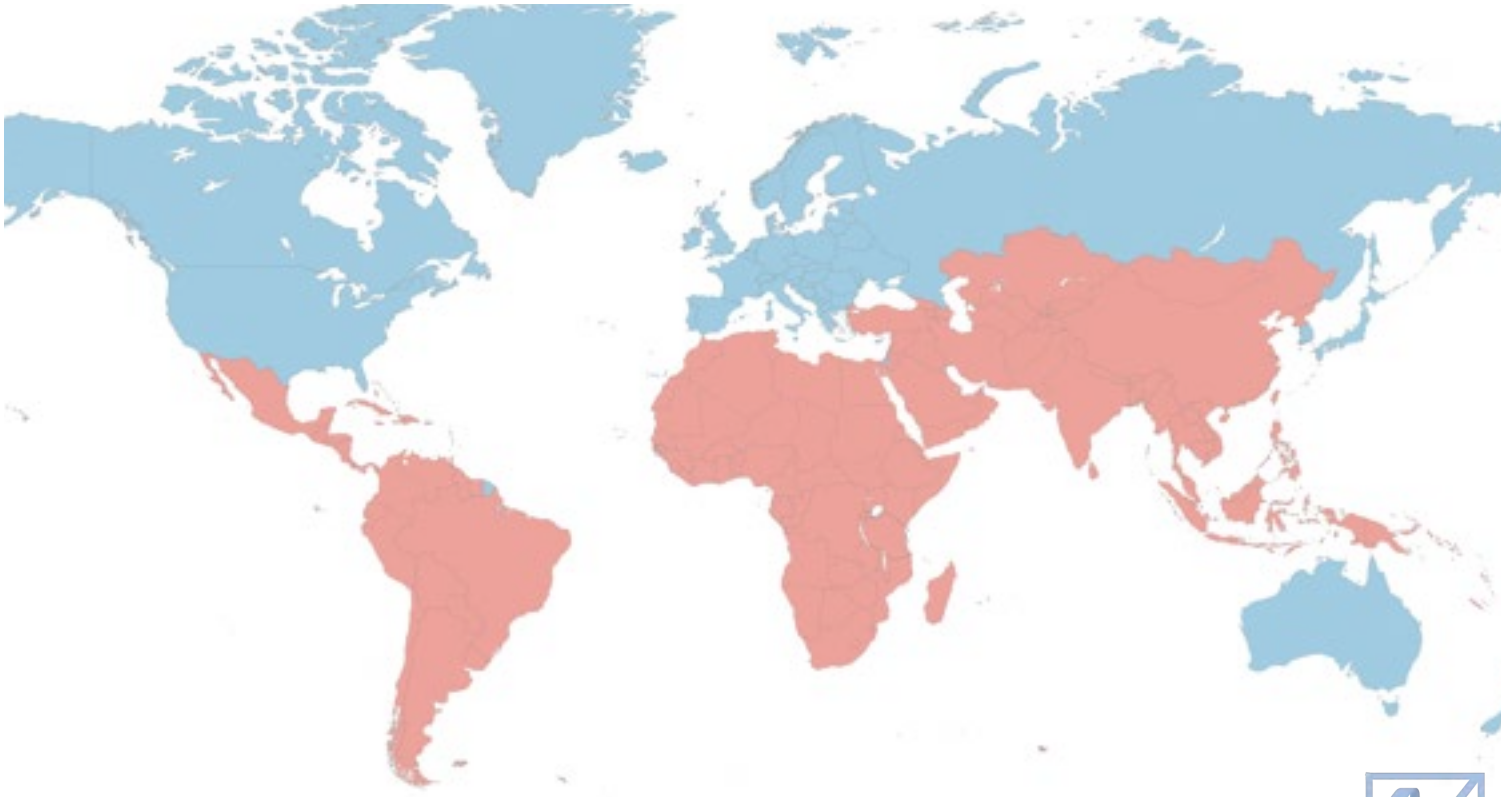

TATARI SYSTEMS

Abstract

This paper looks at whether more dollar-linked crypto tokens have led to cheaper money transfers in developing countries. Where families usually deal with steep charges, slow delivery or shaky local money was an area of focus. When digital dollars cross borders fast and get swapped locally, overall transfer costs drop.

To test this, I use a before-and-after method comparing average fees, in countries like Nigeria and the Philippines, where such tokens are common, versus Vietnam and Ghana, which see less usage despite similar economies and migrant flows. The key result comes from the World Bank's remittance cost index by corridor, grouped by destination country and year between 2011 and 2024.

The methodology exploits cross-country differences in stablecoin adoption and time variation after 2019, using a difference-in-differences design to test whether average remittance fees declined more in high-stablecoin countries than in comparable low-adoption countries.

The starting numbers show that past 2019, Nigeria and the Philippines saw average money transfer costs drop about three points sharper compared to Vietnam and Ghana, down from earlier rates near 8–10%.

A model with country-specific averages shows a slightly weaker negative connection - still noticeable, but uncertain due to data from only four nations. This hints that stablecoin transfers may have reduced costs, however, the finding is preliminary rather than conclusive evidence of causation.

2. BACKGROUND CONTEXT

2.1 Stablecoins, Blockchain, and DLT

Stablecoins run on tech known as distributed ledgers, systems where many computers, or nodes, share an identical history of activity that's hard to fake. No single server runs the show; every machine keeps its own version of the log, updating only when most agree through preset checks. Blockchains, a popular type of this setup, bundle actions into chunks called blocks. One block links to the prior via a digital signature, locking down old info so it can't be changed without notice.

Consensus systems, such as Proof of Work, Proof of Stake, or fault-resistant voting, pick the next valid block. Instead of relying on one central authority, this setup spreads control across nodes, making tampering obvious while letting groups work together despite limited trust. When it comes to payment networks, different blockchain designs bring trade-offs. Open networks allow anyone to join; they're highly accessible plus support modular apps, yet often struggle with slow speeds or fluctuating costs. On the flip side, private versions limit validators to approved members, offering stable operation alongside easier decision-making.

Firmness, when a deal's truly locked in, changes from system to system. Some go with likelihood-based settling, where chances of reshuffling fade gradually. Others give you a set-in-stone result right away. Programs living on the chain, called smart contracts, carry out things like spending caps, holding funds till conditions pass, or needing several approvals before moving value. Because of this setup, coders can launch digital tokens that clear rapidly while sticking to built-in rules anyone on the system checks.

Stablecoins are digital money tied to something steady, often a country's currency, say, the U.S. dollar. Here's how it works: someone gives real dollars to the company behind it; in return, they get digital tokens. Later on, they can swap those tokens right back into dollars whenever needed. As long as people trust that flip-back won't fail, the token holds its peg without drifting much.

Those backed by traditional money stash safe assets, like actual cash or ultra-short government debt, with trusted third-party holders. Plus, updates pop up now and then so folks see proof through checks or financial snapshots.

Crypto-backed stablecoins rely on additional collateral, like shaky digital assets, and dump part of it when values slide. Instead of physical support, algorithmic versions adjust supply steady; yet they've crashed plenty due to weak foundations. In reality, stablecoin transfers run on blockchain; meanwhile, issuing or withdrawing tokens relies on standard banks.

Users move these tokens across wallets just like sending a web link, whereas businesses receive them directly, no credit infrastructure required. Smart functions add safety, for instance, locking money during conflicts, capping usage for inexperienced holders, or cutting off specific accounts to meet legal demands. These digital currencies tie into mainstream finance via supervised trading platforms and payment gateways, while updated protocols now allow retail tools and mobile cash operators to take stablecoins and withdrawals or utility charges.

Emerging markets gain quick payments, transparent costs, plus solid tech backing, each feature paired with simple safeguards. Still, real-world limits influence structure: irregular connectivity points to offline-first designs using QR codes or NFC. Shared devices, or basic phones, need backup methods not tied to one device alone. Cash exchanges rely on linking agents into the system. Gaps in ID checks mean limits should scale by verification level. Clear rules are needed for holding funds and authorizing providers. Overall, DLT builds trust through code, allowing automated functions. Stablecoins act as a known value measure. Smart design choices, in settlement speed, storage setups, access restoration, user guidance on peg risks, turn this base into practical tools.

2.2 Stablecoin Context: Which Are Most Prevalent, Where, and Why?

The top stablecoins now are tied to the dollar, running fast and openly on public blockchain networks. Although USDT leads in transfer activity, USDC builds trust via frequent audits - yet both cut down money movement costs. Their importance comes not from trading gains but practical finance uses. To dodge market swings, investors hold them; families keep value during inflation; companies make quick transfers while skipping traditional banks' delays and fees.

Among these users, cutting costs is a shared goal. Grasping stablecoins is key - since their varied structures influence expenses, access to funds, and consistency in distinct manners.

Dollar-backed stablecoins cut transaction costs by offering steady prices along with wide trading options and strong wallet compatibility - this makes entering or leaving trades affordable. In contrast, alternative stablecoin models tend to bring extra risks, leading to higher expenses via larger bid-ask gaps, unclear redemption terms, or poor usability. Such differences matter when assessing real-world effects: only those stablecoins that consistently swap into local money at minimal expense truly lower payments and transfer charges. Research highlights three key cost-related drivers fueling their use. For one, sending money to emerging markets using legacy channels stays costly; thus, a constantly accessible digital asset moving seamlessly across borders reduces both fees and waiting times.

Second, when prices rise or currencies wobble, families lose money just by keeping cash at home - so coins tied to the dollar cost less to hold, even if they can't reach American banks. Third, new tech companies fit stablecoins into their systems faster than old financial networks, cutting down both regulatory work and daily expenses.

Regional trends show similar challenges. For instance, in the Philippines, stablecoins work alongside mobile payments by cutting cross-border costs. In Nigeria, they aren't only for saving money - people also use them to get paid online at lower expense. Similarly, in countries like Argentina, individuals rely on stablecoins instead of cash dollars to escape fast depreciation of national currencies. In every situation, simple access plus cheap conversion count most - not the tech behind them. Use grows faster when trading sites, payment systems, or online markets offer low-price ways out - since rules only play a role if they change those expenses.

Dollar-pegged stablecoins lead since people value low fees and reliability more than complex tech. Other types backed by crypto emerge only when avoiding control matters more than price, whereas code-based versions seldom handle daily spending. In the end, these coins work best wherever they reliably cut expenses across sending and exchanging money, tying ongoing adoption closely to ease of use, availability, and whether shops take them.

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2.3 Past Innovations Comparable to Stablecoins

Stablecoins fit a broader trend in how payments evolve - each new tool cuts expenses and hurdles tied to shifting funds, even internationally. Past innovations didn't just alter local spending habits; instead, they cut down remittance prices by getting users comfortable with digital money movement while growing affordable access points. Transit cards loaded in advance, along with initial contactless methods, helped buyers and sellers adapt to quick, cheap electronic deals, lessening dependence on physical cash tasks and labor-heavy operations that usually raised transfer rates. Systems like M-Pesa or GCash pushed ahead by slashing charges for sending money at home or abroad via simple mobile setups and widespread agent locations, particularly where banking options were limited.

Research on these models reveals that when depositing and withdrawing cash got easier, transaction fees dropped and more people sent cross-border money, ditching unofficial messengers and costly services.

Fast-payment platforms like UPI or PromptPay made quick, cheap transactions normal by cutting down processing times and operational expenses for banks. Although mainly used within countries, they still pushed legacy money-transfer firms to lower fees due to increased competition - users began expecting faster, clearer service. Stablecoins use similar efficiency tactics yet work beyond national boundaries.

Instead of relying on separate institutional channels, they run on common digital networks; this removes middlemen, speeds up clearing, and reduces currency conversion premiums. They function well with phone-based wallets and local agents, much like older mobile payment models - but unlike those, stablecoins send funds globally without needing correspondent banks, so cost benefits go straight into cross-border cash flows.

3. Hypothesis and Mechanism: Stablecoins and Remittances

Remittances are a natural setting for evaluating the welfare implications of new payment technologies that lower transfer expenses, leaving more money for families in need. As charges drop, a larger share of every remitted dollar reaches recipients in usable local currency, this helps cover basics like meals, education payments, or housing. Speed matters too: instant transfers let households pay utility bills promptly, preventing extra fines or service cuts.

Stablecoins act like a digital link, connecting people in strong-currency nations with those in regions that have less developed financial systems. One user turns U.S. dollars into a token tied to the dollar's value, then sends it via a public blockchain; on arrival, funds are exchanged locally using agents, wallets, or exchanges. This method may lower direct costs from remittance services while also cutting hidden losses caused by poor currency conversion rates. The main idea here is that, post-2019, routes linked to nations with stronger stablecoin use saw steeper drops in typical transfer costs when compared to those tied to places where such digital assets play a minor role.

Nigeria and the Philippines are labeled 'affected' due to their top positions on Chainalysis rankings for crypto activity, alongside widespread access to mobile wallets and physical agents enabling deposits and withdrawals. In contrast, Vietnam and Ghana function as reference points, both get large sums through transfers and share comparable economic conditions, but face tighter rules around digital currencies or show limited integration into everyday spending.

Ideally, tracing the full path, from reduced remittance costs to impacts like school enrollment or owning assets at home, would make sense. Because of limited data, however, this study zeroes in on just the initial part: checking if average transfer fees have changed. People paid less in nations with heavy stablecoin use compared to similar places without it. This hints at better living conditions, but only indirectly: lower costs where stablecoins dominate suggest they might reduce cross-border payment expenses.

4. Data

4.1 Remittance Price Data

The key measure is how much it costs to send money, based on data from the World Bank's Remittance Prices Worldwide (RPW). Every three months, RPW records prices for sending set amounts, either $200 or $500, between particular countries. The total expense includes both direct charges and currency exchange markups, shown as a share of the sum sent. Out of the original RPW entries, I select only those where funds go to Nigeria, the Philippines, Vietnam, or Ghana. Then, for every destination and year, I calculate an average cost across services and routes, resulting in yearly mean rates labeled **avgfee_c_t** between 2011-2024.

4.2 Region Selection Plus Time Frame

The dataset includes 56 country-year cases, four nations across fourteen years. While Nigeria and the Philippines serve as treatment examples, Vietnam and Ghana function as comparison units. As shown in Figure 1, mean remittance charges for both sets decrease prior to 2019; this pattern aligns with the parallel trends premise noted later. Starting in 2019, however, cost reductions seem steeper among treated countries, particularly from 2019 to 2022.

4.3 Macroeconomic controls

To adjust for broader economic factors influencing remittance costs, I include GDP per capita (in current USD) along with consumer price inflation, data drawn from the IMF's World Economic Outlook. These annual figures are matched to the RPW dataset by nation and year. While GDP per capita reflects variations in income and financial system maturity, inflation signals wider economic pressures affecting how service providers set prices. In analyses, these indicators remain in their original scale; given just four countries, complex modeling approaches aren't feasible.

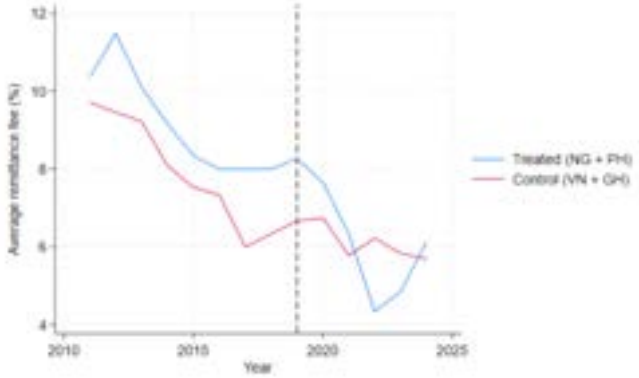
4.4 Stablecoin adoption proxies

Stablecoin information is often incomplete. For 2021–2024, I use regional rankings taken from Chainalysis' worldwide digital currency uptake studies, these include general usage ratings with distinct lists for retail spending, person-to-person transfers, and DeFi use.

In this smaller scope, Nigeria and Vietnam often rank highly worldwide in cryptocurrency uptake, but the Philippines and Ghana show up further down. Still, qualitative data from Chainalysis along with sector analyses indicate that people in Nigeria and the Philippines tend to depend more on USD-pegged stable coins for sending money home or paying across borders, while Vietnamese engagement leans into risk-driven trading. For this reason, I use the Chainalysis figures to support grouping Nigeria and Philippines as strong stablecoin channels compared to Vietnam and Ghana, not as core variables in the primary regression.

4.5 Summary Patterns

Average remittance costs drop from about 10–11% in 2011 to nearly 5–6% by 2024 across all cases. Countries exposed to treatment begin with somewhat elevated fees compared to untreated ones; however, both follow comparable declining paths until the late 2010s. Most pronounced reductions appear after 2019, this timing aligns with a surge in stablecoin usage noted in sector analyses. Such observed changes support the use of a DiD approach outlined later.



This summarizes how intensively the four countries in the sample use crypto assets. The index in the figure inverts Chainalysis' global adoption rank so that higher values correspond to higher adoption. On average from 2021 to 2024, Nigeria and the Philippines score substantially above Vietnam and especially Ghana, indicating systematically higher on-chain activity. Chainalysis' country write-ups also stress that retail use in Nigeria and the Philippines is tilted toward dollar-linked stablecoins and cross-border transfers, whereas Vietnam's activity is more exchange- and trading-driven.

5. Empirical Strategy

5.1 Difference-in-differences specification

To formalize the comparison suggested by Figure 1, I estimate a DiD model at the country-year level. Let avg_feect denote the average remittance fee in country ccc and year t. Define Treatc =1 for Nigeria and the Philippines and 0 for Vietnam and Ghana, and let Postt =1 for years 2019 and later and 0 otherwise. The main specification is:

$$avg_fee_{ct} = \beta_0 + \beta_1 Treat_c + \beta_2 Post_t + \beta_3 (Treat_c \times Post_t) + \gamma_1 GDPpc_{ct} + \gamma_2 Inflation_{ct} + \varepsilon_{ct},$$

estimated using ordinary least squares with standard errors clustered at the country level. The coefficient of interest is **β3**, which measures the additional change in remittance fees in treated countries after 2019 relative to the change in control countries. Because unobserved country characteristics may affect fee levels, I also estimate a fixed-effects version that absorbs time-invariant country heterogeneity:

$$avg_fee_{ct} = \alpha_c + \lambda_t + \beta_3 Post_t + \beta_3 (Treat_c \times Post_t) + \gamma_1 GDPpc_{ct} + \gamma_2 Inflation_{ct} + u_{ct},$$

where **αc** are country fixed effects and **λt** are year dummies. In practice the year dummies are absorbed by the Post indicator, since the panel is small, but the fixed-effects estimator still exploits within-country variation over time.

The treatment definition and the choice of 2019 as the post period are therefore guided by both the adoption evidence in Figure prior and the narrative in Chainalysis reports that place stablecoin-based remittances and cross-border payouts at the center of usage in Nigeria and the Philippines.

5.2 Identification and limitations

The main assumption is without stablecoin payment systems, all four nations would've seen alike shifts in remittance costs post-2019. As shown in **Figure 1**, a graphical review reveals fee reductions progressed similarly across groups before 2019, supporting matching patterns early. Adding GDP per capita along with inflation adjusts for broader economic disruptions that shift prices unevenly. Meanwhile, the setup has notable drawbacks. To begin with, just four nations are covered in the data, making regression results with grouped errors quite weak. Next, exposure is assessed per nation through ranked and interpretive signs of stablecoin use instead of actual transaction records from blockchain transfers. Then again, post-2019 includes COVID, shifts in financial oversight, and broader access to digital wallets, factors that might affect costs together with or apart from stable assets. Because of this, findings reflect observed links aligned with how stablecoins could work, yet don't confirm cause-and-effect outcomes.

6. Results

6.1 Baseline difference-in-differences estimates

Table 1 shows the basic regression linked to equation (1). For treated nations during the time after 2019, the interaction coefficient stands at -3.14; its clustered standard error is 0.83, this gives a t-value of -3.8 along with a p-level of 0.03. The result suggests that following this point, outcomes shifted downward by roughly 3.1% on average. In 2019, Nigeria and the Philippines saw remittance fees drop around 3.1% further compared to Vietnam and Ghana, adjusting for GDP per person and price changes. With initial charges near 8–10%, this decline means costs fell by approximately a third in treatment countries versus comparison ones. The main effect estimates support this. Although the treatment is positive, around 1.2%, it suggests treated nations had higher charges than untreated ones. While the post-period indicator is negative,−0.7%, it reflects a minor drop in fees among untreated countries after 2019. Because the model accounts for half the fluctuations across country-year, pairs remain unexplained.

Linear regression

Number of obs =	56
F(2, 3) =	.
Prob > F =	.
R-squared =	0.4881
Root MSE =	1.5004

(Std. err. adjusted for 4 clusters in ctry)

avg_fee	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
treat	1.195693	.5918789	2.02	0.137	-.6879302	3.079316
post	-.7059267	.4098303	-1.72	0.183	-2.05819	.5983364
did	-3.143831	.8320136	-3.78	0.032	-5.79087	-.495193
gdp_pc	-.0013876	.0006053	-2.16	0.120	-.0032338	.0006186
inflation	.011134	.0217392	0.51	0.644	-.0580497	.0803178
_cons	11.06211	1.14721	9.64	0.002	7.411181	14.71305

6.2 Fixed-effects specification and robustness

Column (2) in Table 2 shows results using fixed effects. When country-specific factors are accounted for, the interaction coefficient stays negative, specifically −2.32%, which is sizable in real-world terms; however, the clustered standard error increases to 1.30, leading to a p-value of 0.17. Given there are just four clusters, reduced accuracy isn't unexpected.

Fixed-effects (within) regression

Group variable: ctry

Number of obs =	56
Number of groups =	4

R-squared:

Within = 0.4678

Between = 0.1400

Overall = 0.3564

Obs per group:

min =	14
avg =	14.0
max =	14

F(3, 3) =	.
Prob > F =	.

corr(u_i, Xb) = 0.0117

(Std. err. adjusted for 4 clusters in ctry)

avg_fee	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
post	-1.012395	.707918	-1.43	0.248	-3.265306	1.240516
did	-2.325503	1.304157	-1.78	0.173	-6.471913	1.820907
gdp_pc	-.0008893	.0007381	-1.20	0.315	-.0032382	.0014596
inflation	-.0192054	.0384879	-0.50	0.652	-.1416912	.1032803
_cons	10.59477	1.661327	6.38	0.008	5.307685	15.88185

These findings still align with the main argument: treatment countries saw steeper fee drops compared to controls, yet limited data weaken definitive claims.

6.3 Graphical evidence

Figure 1 adds context to the regression results by showing fee trends for both treated and control groups from 2011 to 2024, using a dotted line to mark 2019. Before 2019, fees in both groups fall at roughly the same rate, supporting the idea of parallel paths. Following 2019, however, the drop accelerates in the treated group, especially through 2022, while the control group keeps its steadier decline. The graph's shape reflects the negative coefficient found in models; it also supports the claim that routes with heavier stablecoin activity saw steeper drops in costs.

7. Discussion

The data show lower remittance costs in Nigeria and the Philippines post-2019 compared to Vietnam and Ghana. This trend matches reports indicating stablecoins gained traction in sending money within Nigeria and the Philippines at that time, yet played a smaller role in Vietnam and Ghana. Put differently, when fees dropped most was also when digital dollar tokens began spreading, hinting they may reduce cross-border transfer expenses. Meanwhile, stablecoins aren't the sole new factor shaping these routes. Instead, mobile payments, digital remittance platforms, or real-time transaction networks also grew during the study window. In turn, such shifts may have helped each of the four nations. However, the diff-in-diff setup aims to capture trends unique to heavy stablecoin adopters; still, it can't fully untangle stablecoins from other policy advances more common in Nigeria and the Philippines. Take, say, rules that push organizations to act, yet compliance isn't automatic despite external demands Fees on classic money transfers, or pressure from online-only companies, hit harder in these areas.

The small sample size plays a role. Using just four nations along with fourteen years makes results depend heavily on how treated groups are set, as well as when the post-treatment phase begins. While the main setup shows an interaction effect that's statistically meaningful, this vanishes under fixed effects once individual country levels are accounted for. Given there are only four clusters, such an outcome isn't unexpected, still, it suggests figures should be seen as hints, not exact values. More robust conclusions could emerge from broader data, including additional remittance-reliant countries and extended periods of stablecoin adoption.

Even with these limits, the study adds to growing talks about how stablecoins actually affect economies. Public conversation often centers on speculation or fears they could replace bank savings. This research highlights a more specific, practical use: cutting costs and speed bumps in routes where workers frequently transfer funds abroad. When stablecoins are used mostly to dodge high fees and slow transfers, not to gain profits, the impact looks less like digital asset trends and more like earlier studies on sending money across borders.

The supplement outlines initial models for predicting uptake; however, the core contribution remains clear evidence of a consistent cost structure tied directly to cross-border payments via stablecoins.

8. Conclusion

This study aimed to examine a simple idea: in nations where dollar-linked stablecoins are commonly adopted for international transfers, typical remittance costs would decline more quickly compared to those without such usage. For nations like Nigeria, the Philippines, Vietnam, and Ghana, where stablecoins play a smaller role, I used World Bank RPW figures from 2011 to 2024. Alongside this, IMF economic indicators were included; Chainalysis adoption scores also informed the model.

A basic difference-in-differences approach was applied instead of more complex methods. Results show that post-2019, these countries saw about 3% lower remittance costs compared to peers, beyond general declines seen across regions. While fixed-effects models support similar conclusions, their confidence intervals are wider.

This doesn't confirm stablecoins led to lower fees. Data comes from a limited set, hard to isolate stablecoins from other digital payments. Yet, the speed and size seem unlikely if stablecoins had zero impact. Rather, results align with the idea that they act as an added level within wider payment systems, reducing costs when used alongside mobile wallets, agents, or open trading venues. To policymakers, the study suggests real advantages of stablecoins might show up in everyday use, not just high-profile trading.

Where authorities assess dangers tied to dollar-pegged digital currencies, they should account for how they may lower transfer costs for families. Researchers are reminded that improved information is needed, linking blockchain with pricing trends and family-level effects.

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NATIONAL SECURITY LANDSCAPE OF GPUS

Global AI growth is turning GPUs into strategic infrastructure, with demand and shortages driving pre-leased data center capacity and making GPU access a national-power issue rather than just a commercial one. Ethiopia's "Digital Ethiopia 2025" push is expanding digital services, telecom reach, PKI/e-signature systems, and early data-protection rules, but the country still relies on imported GPUs and turnkey data-center modules, leaving it exposed to FX constraints, customs delays, export controls, and vendor/geopolitical dependence (often China-linked supply). That dependency collides with hard operational realities: high-density GPU clusters need reliable power and cooling, stronger physical and cyber security, and scarce specialized talent, while shifting government posture on crypto mining shows policy risk can reprice the whole market quickly.

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NATIONAL SECURITY LANDSCAPE OF GPUS

The Global GPU Landscape

The rapid expansion of artificial intelligence (AI) has driven unprecedented demand for both data centers and graphics processing units (GPUs). GPUs are crucial for high-performance computing, enabling large-scale data processing, graphics rendering, machine learning, and blockchain-related compute tasks (use of gpu). Reflecting this surge in demand, the global GPU market was valued at approximately USD 18.4 billion in 2024 and is projected to reach USD 92 billion by 2030 (econ stats). Governments and private-sector firms are investing heavily to secure access to GPU capacity and the data-center ecosystems required for advanced operations (how gpus are wanted).

The United States remains the dominant global market, followed closely by the Asia-Pacific, led by China, South Korea, and Japan. Europe has lagged; however, forthcoming regulatory features are set to increase demand for GPUs. African participation remains limited, though gradually growing, with initiatives to develop the continent's first "AI factory" dedicated to AI computing (south african ai factory). The effort has already begun in South Africa, with planned expansions to Egypt, Kenya, Morocco, and Nigeria in the future, collectively signaling an emerging regional demand for GPU resources (expansion of ai facts).

Persistent supply shortages have contributed to the emergence of pre-lease data center capacity. While smaller entities struggle to face high barriers to entry, data centers have become a more compelling investment opportunity for emerging digital economies seeking to develop foundational AI infrastructure. It will analyze Ethiopia's political and regulatory environment, GPU-security-related issues, potential supply-chain risks, the strategic value of GPUs in a national context, and geopolitical dynamics' effects on GPU control. The focus is within 2020–2025, with future projections looking to 2030.

The geographic scope centers on Ethiopia, with comparative references to South Africa, along with specific East African nations. Key business sectors analyzed will include crypto mining, telecommunications, national defense/security, GPU hardware, and data centers. The analysis will include supply-chain organization, policy review, vulnerabilities and potential risks, comparative analysis, and scenario considerations.

Ethiopia's National GPU Landscape

Ethiopia's pursuit of digital sovereignty and transformation of its digital identity is anchored in its "Digital Ethiopia 2025" initiative. Approved by the Council of Ministers in 2020, the strategy seeks to accelerate Ethiopia from an agrarian economy into the digital era (Digital Ethiopia 2025: Progress, Challenges, and the Road Ahead). The initiative is structured around four pillars: physical infrastructure, enabling systems, digital applications, and broader ecosystem development. Its implementation has supported growth in industries such as digital banking, the digitization of hundreds of government services, and the rise of early-stage digital firms (same as one before).

As of early 2025, Ethiopia had approximately 28.6 million internet users, around 21.3% penetration of its nearly 134 million population (Digital 2025: Ethiopia — Data Reportal – Global Digital Insights). Mobile connectivity exceeds this, with around 85 million links (~63.8% of the population), highlighting a predominantly mobile-based and dispersed digital-access landscape (same as one before).

Ethio Telecom dominates Ethiopia's telecommunications infrastructure, with fiber networks through Sudan, Djibouti, Kenya, and Somaliland (Ethiopia - Digital Economy). Market reports increasingly identify Ethiopia as a favorable market for future data center growth due to supportive digital policy and localization needs.

Ethiopia has also begun strengthening its digital security infrastructure through the launch of a national Public Key Infrastructure (PKI) at the Information Network Security Agency (INSA) headquarters (Ethiopia Launches National Public Key Infrastructure - ID Tech). The PKI establishes centralized issuance of digital certificates and supports secure communications for banking, emails, VPNs, and software distribution through encryption. This builds on Ethiopia's Electronic Signature Proclamation (beginning in 2018), which established legal recognition for electronic signatures, online contracts, and electronic records.

The regulatory framework was further refined in 2020 through standards governing electronic signatures. In 2024, amendments allowed governing bodies to regulate and use directives to implement the law more effectively. The regime mandates an asymmetric cryptosystem to protect electronic signatures and prevents private entities from issuing certificates, with INSA oversight. These initiatives have supported Ethiopia's national digital identification program, with up to 24 million Ethiopians enrolled to date.

The Personal Data Protection Proclamation of 2024 established safeguards related to consent, data breaches, confidentiality, and constraints on cross-border transfers (Ethiopia's Personal Data Protection Proclamation of 2024 and its Budding Digital Identity Regime – CIPIT). While the law remains underdeveloped, it marks Ethiopia's transition toward stronger data-protection regulation. Additionally, the Digital Ethiopia strategy acknowledges the need for more specific digital-competition laws, which have not been revised since the 2013 proclamation.

From a financial perspective, Ethiopia's fintech sector has emerged as a successful beneficiary of Digital Ethiopia's reforms. The Payment Instrument Issuers' Directive (2020) accelerated the adoption of mobile financial services, contributing to a 129% increase in digital payments last year, even with underdeveloped infrastructure.

Fintech remains a strategic focus as Ethiopia advances toward 2030 (Ethiopia's approach to their National Data Strategy offers lessons for Africa's digital future). Complementing these efforts is an initiative to develop a nationwide business portal integrating services from ten government agencies, including automated business licensing. Implementation challenges remain due to system maintenance. Efforts to grow digital literacy through the "5 million coders initiative" have shown limited progress so far (Digital Ethiopia 2025 Strategy Boosts Global Competitiveness: Ministry - ENA English), though the movement is expected to continue through a 2030 extension and improve as younger cohorts enter the workforce.

While digital-economy activity has expanded across sectors, growth remains constrained by infrastructure limitations. Firms with reliable connectivity have demonstrated stronger scalability and user growth, reinforcing the importance of stable infrastructure and reliable GPU supply chains (Road to Digital Ethiopia: Why Reliable Infrastructure Is Critical to Vision 2025).

Looking toward 2030, Ethiopia's growth will require sustained funding. GPUs are expected to play an irreplaceable role in enabling Ethiopia's growth, reinforcing the need to attract and leverage foreign investment, similar to peers such as Kenya and Rwanda. Overall, while Digital Ethiopia 2025 has been transformative, its limitations highlight national security risks related to dependency on foreign investment and trade, infrastructure gaps, human-capital shortages, and connectivity constraints.

Infrastructure and Usage:

GPU production is complex, requiring sophisticated manufacturing and hard-to-obtain components. This means countries like Ethiopia, aiming to enter the AI industry, must import GPUs.

For Ethiopia, this has created a unique opportunity: combined with low-cost energy, it has positioned the country as a bitcoin mining hub due to low operational costs and energy policies that generally support high data-processing operations (Ethiopia Energy Intensive Tech opportunities). Ethiopia formally opened to Bitcoin mining in 2023, attracting entities to develop data centers despite restrictions by the National Bank of Ethiopia on cryptocurrency as legal tender.

As of August 2025, Ethiopia has begun shifting away from purely bitcoin mining toward AI training operations (ethiopian feezing mining). Public pressure and grid constraints have contributed to the phase-out of cryptocurrency mining, with no new contracts expected and existing agreements under review. Ethiopian Electric Power supported initial adoption as a way to generate foreign currency (reportedly bringing in USD 220 million) (Ethiopia Intends to Wind Down Bitcoin Mining Citing Grid Strain).

Despite these shifts, Ethiopia signed a USD 250 million agreement with a Hong Kong-based firm in February 2024 to support bitcoin mining operations (250 m investment). In parallel, Huawei Digital Power has emerged as a key provider, deploying AI-enabled data center technology at the Cooperative Bank of Oromia in Addis Ababa—described as Sub-Saharan Africa's first AI-integrated data center power deployment. The systems include Huawei's FusionModule2000 and UPS5000-E, integrating AI-based inference and monitoring hardware (Huawei and Cooperative Bank boost digital finance in Ethiopia with green data centers | Extensia Ltd). Ethiopian distributors list AI-integrated modules at USD 200,000–2 million, depending on the GPUs installed and configuration (same as the one before).

Collectively, these developments highlight Ethiopia's ambition to expand into AI while maintaining regulation and protecting national interests.

Digital Sovereignty and Security

Control over global GPU supply has direct implications for Ethiopia's AI trajectory. The United States and China remain dominant GPU exporters, shaping access through trade policy and geopolitical motives (what can affect crypto).

Ethiopia's GPU imports are primarily sourced from Chinese companies, supplemented by the UAE, India, and France. China has strengthened economic relations with Ethiopia for years through more accessible trade and investment (China-Ethiopia industry matchmaking event held in Addis Ababa to boost BRICS cooperation - Global Times). Huawei Digital Power is a key Chinese investor, collaborating with Cooperative Bank of Oromia (COOP) and helping address infrastructure design constraints (Huawei power company investment). COOP, one of Ethiopia's largest banks with over USD 25 billion in transaction volume, required a more power-efficient and scalable structure—an issue Huawei Digital Power addressed. This illustrates the upside of foreign investment in closing infrastructure gaps, while also underscoring Ethiopia's dependency on external suppliers.

From a national security perspective, several risks and screening needs follow from this dependency:

- No domestic semiconductor capacity: Ethiopia lacks domestic semiconductor manufacturing and GPU design capabilities; racks, cooling modules, GPUs, and high-end server components must be imported. This increases exposure to supply bottlenecks, FX volatility, customs delays, and potential embargoes.
- Infrastructure constraints for GPU clusters: High-density GPU clusters require reliable power and robust cooling. Outside urban centers, grid reliability and limited cooling capacity raise operational risk. Diesel backup can mitigate outages but introduces cost, logistics, and resilience tradeoffs.

- Data center retrofits: Many existing Ethiopian data centers may not meet GPU-cluster specifications. Scaling will require capital-intensive upgrades or purpose-built facilities.
- Regulatory scrutiny and security risk: Imports may face inspection by bodies such as INSA or ICT, and sensitive compute tied to foreign data flows may face stricter oversight. Physical risks (theft, sabotage) and technical risks (backdoors) remain concerns.
- Human capital constraints: Limited availability of system and GPU engineering talent increases reliance on foreign expertise, which can create operational and security dependencies.

Despite these risks, the environment presents investor opportunity: the market is underdeveloped, creating first-mover upside for firms that can manage infrastructure and regulatory complexity. Tatari could potentially capture regional demand through a data-residency advantage and by aligning with national initiatives like Digital Ethiopia 2025 to pursue contracts with emerging AI programs.

Comparative Analysis

Market research estimates South Africa's data-center market at around USD 2.16 billion in 2024, projected to reach USD 3.4 billion by 2030 (southafrica datacenter stats). South Africa's data-center GPU market generated approximately USD 120.4 million in 2024 and is projected to grow to USD 1.6765 billion by 2033, implying a 36.8% CAGR (gpumarket). Growth has been driven by major AI/GPU investments from Nvidia, Cassava, and Microsoft (same link as the one before + microsoft investment).
This momentum has been supported by clearer policy frameworks around crypto, AML/CFT oversight, and CASP licensing (aml/cft + casp + crypto laws), alongside stronger data sovereignty rules and data-protection laws.

By contrast, Ethiopia's less developed crypto and data-protection regimes, combined with lower levels of private investment and continued reliance on external hardware supply, slows acceleration of a mature GPU–data-center ecosystem.

National Security Considerations

Ethiopia's digital ambitions face operational and strategic constraints:

- Power and cooling as core vulnerabilities: GPU operations depend on reliable energy and cooling; outages and thermal failures can quickly disrupt clusters. Risks increase outside urban areas due to weaker grid resilience, environment (heatwaves), and water constraints.
- Supply-chain disruption exposure: Import dependence creates vulnerability to export controls, geopolitical blockades, bottlenecks, and hardware tampering.
- Underdeveloped verification capacity: Sensitive workloads require robust scanning, validation, and integrity checks for imported hardware—capabilities that remain limited.
- Cyber and physical threat surface: As GPU clusters become strategic assets, sabotage and cyber threats become more material, requiring stronger national security oversight and incident response.

Key takeaways include power/cooling threats, dependence on foreign hardware and personnel, and data oversight/security, each critical when expanding in Ethiopia.

Ethiopia's Strategic Landscape & Future Outlook

Building on the analysis above, two illustrative scenarios help frame Ethiopia's trajectory and Tatari's operating environment.

Case 1: Ethiopia as a GPU Hub

This assumes Ethiopia successfully scales GPU data centers across major regions, with the government playing a pivotal role through improved licensing, upgraded utilities, and stronger backup power capacity. This could enable domestic GPU rental markets and potentially regional exports of compute services. National outcomes would include strengthened digital sovereignty, expansion of government services (ID systems), and faster private-sector development.

Prerequisites: improved regulatory policy, targeted government support/subsidies for tech sectors, and international partnerships.

Case 2: Ethiopia grows but remains contained

This scenario assumes slow policy evolution and persistent infrastructure constraints, compounded by supply-chain delays, licensing friction, and weaker investment momentum. These constraints would deter major clients and investors. For Tatari, growth would skew toward domestic demand only, while better-positioned regional competitors could dominate cross-border and export-oriented compute markets, narrowing the window for regional expansion.

Prerequisites: policy development and limited government investment in critical infrastructure.

High-Performance Computing (HPC) running CPU and GPU nodes

High-Performance Computing (HPC) clusters refer to computationally intensive systems built by combining multiple interconnected computers to run large workloads. These clusters can include different types of compute nodes: CPU-only nodes for complex sequential tasks, GPU-accelerated nodes for massively parallel calculations, or hybrid configurations that leverage both. Traditional data centers are typically composed of standard compute and networking architecture, with CPUs handling core workloads such as networking, applications, and storage. By contrast, AI-oriented data centers rely more heavily on advanced accelerators such as GPUs and Tensor Processing Units (TPUs). Major tech companies like Meta are exploring TPUs to diversify their infrastructure, since many deployments currently depend primarily on Nvidia GPUs.

High Performance Computing (HPC) Clusters combine multiple interconnected computers (nodes) to solve computationally intensive problem		
CPU (Central Processing Unit) CPU-intensive tasks are operations that require a lot of processing power to complete, often involving complex calculations, data processing, and heavy computations.	**GPU (Graphics Processing Unit)** GPU-intensive tasks are those that involve parallel processing, such as gaming, 3D rendering, and artificial intelligence (AI) and machine learning (ML) model training. (By NVIDIA)	**TPU (Tensor Processing Unit)** TPU tasks are those that involve tensor operations and matrix multiplications, such as training large language models, neural network inference, and deep learning research. (By Meta)
Few powerful cores (typically 4-64) optimized for complex, sequential tasks	Thousands of smaller, simpler cores optimized for parallel processing	Contain specialized matrix multiplication units (MXUs) rather than traditional cores
Typical applications: • Complex logic and decision-making-branching code with lots of "if/then" statements • Serial processing - tasks that must be done step-by-step • Low-latency operations - quick response times for individual tasks • General-purpose computing varied workloads	**Typical Applications:** • Massive parallelism - same operation on lots of data simultaneously • Mathematical computations - especially matrix operations • Data parallelism - when you can split work into independent chunks • Training and Inference	**Typical Applications:** • Training large language models • Neural network inference at scale • Computer visions tasks • Scientific computing and research • Generative AI applications • Speech recognition and synthesis

Advanced Cooling Systems

Data center temperature must be maintained within a defined range to prevent condensation and damage to IT equipment, which can cause downtime and data loss. Energy efficiency is also crucial. Modern AI data centers aim to achieve strong power usage effectiveness (PUE), an efficiency metric used in the data center industry to quantify how much total facility energy is spent supporting IT equipment. The closer PUE is to 1.0, the more energy-efficient the data center is. An "ideal" PUE typically falls in the 1.1–1.3 range, meaning that for every 1 watt consumed by computing equipment, only 0.1–0.3 additional watts are consumed by cooling and other supporting infrastructure. According to the Uptime Institute, the global avg. PUE was 1.56 in 2023.

$$PUE = \frac{\text{Total Facility Load (cooling network, lighting, IT load...)}}{\text{IT Load}}$$

often in ways that are incompatible with traditional cooling mechanisms. As rack densities rise—especially in the 50–100 kW per rack range—more sophisticated cooling systems become necessary (Cai & Gou, 2024). Cooling-related energy costs can represent 30–40% of total operating expenses (Capozzoli & Primiceri, 2015). A Microsoft study found that cold plates and two immersion-cooling technologies reduce greenhouse gas emissions by 15–21% over their life cycles, reduce energy demand by 15–20%, and reduce water consumption by 31–52% in data centers, compared with air cooling (Microsoft, 2025). As compute requirements increase, cooling systems typically need more advanced liquid-cooling designs and fluids. Figure A demonstrates how rising workload energy demands increasingly point toward liquid cooling as a reliable method.

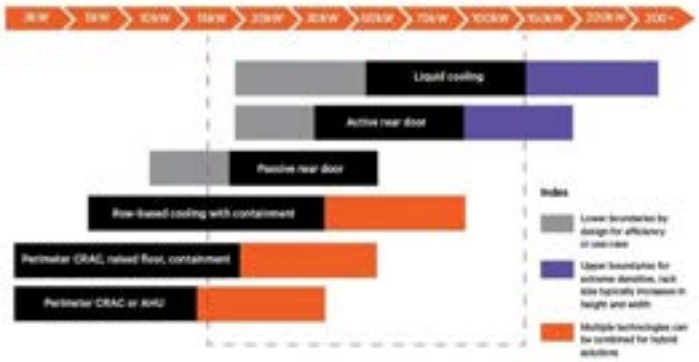


Fig A. How cooling technology evolves as rack density increases in the data center, where red boundary highlights conditions under which liquid cooling is a viable option.

Liquid cooling methods leverage the thermal transferring properties of water or dielectric fluids: these dissipate heat and are 3000 times more effective than air cooling alone (Vertiv, 2025). Data centres utilizing both air and liquid cooling can reduce Total Cost of Ownership (TCO) compared to solely air- cooled systems (Sorell et al., 2015). Trends indicate a gradual shift towards including liquid cooling: around 17% of data centres use liquid cooling, and over 61% are considering its adoption, showing a strong shift toward this approach (Uptime Institute, 2025).

Direct to Chip Cooling and Precision Cooling:

Direct-to-chip cooling is a technique in which pumps circulate a dielectric fluid through tubing to cold plates mounted on chips. The fluid absorbs heat at the source and is then routed away for heat rejection. Because it removes heat directly where it is generated, direct-to-chip cooling typically delivers superior performance compared with traditional air cooling. Notable innovators in this space include Chilldyne and JetCool.

Precision cooling systems are designed to remove heat directly at high-performance computing components rather than cooling entire rooms. These systems deliver chilled air or liquid coolant precisely where thermal loads are highest, typically using in-row or in-rack cooling units with sensors and control systems that monitor temperature in real time.

They dynamically adjust airflow or coolant flow rates to match a server's thermal output. Advanced deployments also use containment strategies such as hot-aisle/cold-aisle configurations to prevent mixing of exhaust and intake air, improving cooling efficiency. Some solutions integrate liquid-cooling loops connected directly to server racks and use heat exchangers to transfer heat from IT equipment into facility chilled-water systems.

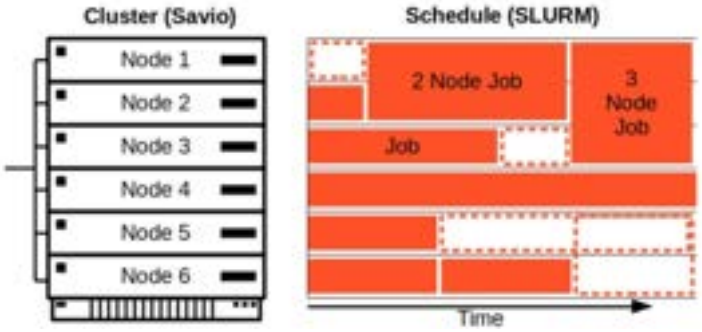
The market for liquid cooling is expanding rapidly, growing from roughly $100–150 million in 2020 to a projected $1.7 billion by 2027—more than a 10× increase over seven years. Revenue is often cited as approaching $2 billion by 2027, implying a roughly 60% CAGR from 2020 to 2027. By 2027, direct liquid cooling is projected to account for approximately 60–65% of the overall market. Major suppliers include Vertiv, Schneider Electric (APC), STULZ, Rittal, and Asetek.

Memory Management and Network Bandwidth

Network bandwidth measures how much data can move between servers, storage systems, and external networks per second—typically expressed in Gbps (gigabits per second) or Tbps (terabits per second). If an AI workload starts with modest compute needs but later scales to require 100× more resources (for example, by expanding dataset size), the data center must be able to accommodate that growth without becoming bottlenecked.

Large language models with billions of parameters require very high high-bandwidth memory (HBM) capacity and throughput to store model weights, intermediate calculations, and training data. Meta has documented approaches to reduce "stranded" memory and platform overhead (the "memory tax"), including right-sizing, compression, and offload mechanisms. For example, Meta reports that Transparent Memory Offloading has been in production since 2021 and saves approximately 20–32% of memory per server across millions of servers, including an average ~9% reduction in total server memory (Meta, 2022).

On the hardware side, HBM3 can deliver sustained memory throughput exceeding several terabytes per second at the system level. Newer HBM3E products push further: SK hynix states its HBM3E can process up to ~1.18 TB/s per stack, while Micron describes HBM3E bandwidth of "more than 1.2 TB/s per placement" and advertises ~30% lower power consumption than competitors for its HBM3E offering (Micron, 2025).



A SLURM job allocation visualization graph

SLURM is widely used by cloud providers, manufacturers, AI companies, and research labs across industries such as autonomous driving, healthcare and life sciences, energy, financial services, manufacturing, and government. It is reportedly used in more than half of the top 10 and top 100 systems in the TOP500 list of supercomputers—including systems like Frontier, Polaris, and Perlmutter (NVIDIA Blog/WebWire; Nvidia, 2025). Kubernetes is a common alternative, but it generally prioritizes high availability and service orchestration over pure scheduling efficiency.

For organizations running diverse workloads—especially those combining training and inference services—or operating in cloud-native environments, Kubernetes may be more appropriate despite its increased complexity.
In practice, the choice often depends on the workload mix (batch training vs. always-on services), the complexity of client demands, and whether the organization is optimizing primarily for utilization/scheduling (often SLURM) or availability/operational orchestration (often Kubernetes).

Hardware Example:
AI operations require seamless coordination across heterogeneous computing resources—from GPU-accelerated servers running training jobs, to storage systems housing massive datasets, to network devices enabling high-speed data transfer. A management platform can be highly effective in provisioning, automating, and controlling diverse hardware components (such as servers, virtual machines, network infrastructure, and power distribution units), allowing IT teams to deploy and reconfigure resources as AI workloads shift between compute-intensive training phases and latency-sensitive inference deployments (e.g., LLM training vs. inference). One example of this type of platform is Vertiv's Avocent DSView solution.

Management Platform:
A management platform enables secure remote monitoring and control of IT infrastructure, helping infrastructure teams automate routine processes and streamline expanding operational responsibilities.

Rack Manager:
A rack manager physically connects to interface modules—or directly to service processors, rack PDUs, or network equipment. It can provide power over Ethernet (PoE), IP consolidation, and network address translation (NAT) to simplify and secure access across the rack environment.

Interface Modules:
Interface modules connect aggregation appliances to end targets. These IP-based devices are typically either KVM modules or serial modules, enabling remote access and management for a wide range of equipment types.

This architecture supports resilience against outages, which are common in remote areas and can be critical from a downtime perspective. For example, some deployments use an always-connected, out-of-band management platform (such as the Avocent MP1000) that can recover from outages via cellular connectivity. This makes it possible to maintain access to customer equipment and restore service even when Ethernet is down—or not available at all.

Overview of Vertiv™ Avocent® DSView™ Solution

Advertised benefits include:

- Deliver IT services rapidly and reliably through automation, integration and remote access.
- Detect and diagnose incidents faster with real time insights into health and status.
- Protect and shield IT devices using enterprise grade secure platforms.

FEATURED FOUNDING PARTNERS

1 Al-Amin Mumuney

Role: Head of Strategic Partnerships
Hometown: Abuja, Nigeria
Prior Relevant Roles:
Nigerian Stock Exchange, Estu

2 Amen Amare

Role: Operating Partner, CEO
Hometown: Addis Ababa, Ethiopia
Prior Relevant Roles:
Energy Private Equity, Apple Finance, Goldman Sachs Markets (Digital Assets)

3 Meba Michael

Role: Chief Financial Officer
Hometown: Addis Ababa, Ethiopia
Prior Relevant Roles:
African Private Equity, Renewables Private Equity

4 Yasha Genkin

Role: Chief Strategy Officer
Hometown: New York City, NY
Prior Relevant Roles:
Cover'd, Paggi Physical Therapy, Reuth

EXECUTIVE LEADERSHIP

1 Christian Assefa

Role: Head of Investor Relations

Hometown: St. Louis, MI

Prior Relevant Roles:
Equity Analyst at BLK Capital Mgmt., Consultant at Local Initiatives Support Corporation

Qualified Impact Highlight:
Helped ensure the company's securities achieve fair valuation by accurately communicating financial performance, strategic goals, and market insights.

2 Glodi Karagi

Role: Head of Business Development

Hometown: Worcester, MA

Prior Relevant Roles:
Investment Fellow at Roark Capital, VC Analyst at igniteXL Ventures and PE Intern at Project Destined

Qualified Impact Highlight:
Originated and proposed BTC-backed SPV (Structured Product Vehicle) derivative offering for Tatari's network

3 Jaemoon Lee

Role: Chief Location Strategist

Hometown: Providence, RI

Prior Relevant Roles:
Defense Health Sub-Contracting & Government Finance Posts

Qualified Impact Highlight:
"Single source of truth" decision systems fast: took messy, multi stakeholder finance inputs and turned them into a reconciled model executives could actually run with

4 Jimi Oso

Role: Chief Engineering Officer

Hometown: Lagos, Nigeria

Prior Roles: Expedia ML, Massachusetes Datacenters

Qualified Impact Highlight:
Facilitated procurement and feasibility study of Reference Design Architectures

5 Lidya Demessie

Role: Chief Operations Officer

Hometown: Silver Spring, MD

Prior Relevant Roles: Fellow at Roark Capital and Equity Analyst at BLK Capital Management

Qualified Impact Highlight:
Overseeing operations, ensuring efficiency through managing HR and finance

6 Nathan Banketa

Role: Head of Research

Hometown: Dallas, TX

Relevant Roles: Blackrock Aladdin Data Ops., & Enterprise Systems Interat Takeda

Qualified Impact Highlight:
Led and edited all Tatari Reports, Country-Deep Dive and oversaw company back-end infrastructure

OPERATING TEAM

1 Abigail Melkamu

Role: Growth & Operations Intern
Hometown: Atlanta, GA
Prior Relevant Roles:
Search fund, Research with Nobel Laureate on economic distribution, worked for an Angel Investor Group

2 Cale Carpenter

Role: Growth & Operations Intern
Hometown: Colorado Springs, CO
Prior Relevant Roles: VP of one of ASU campus's most prestige finance clubs - current M&A IB intern at a local Phoenix bank

3 Darley Rene

Role: Growth & Operations Intern
Hometown: New York, NY
Prior Relevant Roles: Federal Reserve Bank of New York, Project Destined Private Equity Intern

4 Liz Ramsay

Role: Strategic Capital & VC Outreach
Hometown: Los Angeles, CA
Prior Relevant Roles: Professional Modeling, Magazine Editor for NYU Luxury & Retail Association, Editor for Frontrunner Magazine & PR Intern for Green Valley Project

5 Mahak Reji

Role: Strategic Capital Intern
Hometown: Dubai, UAE
Prior Relevant Roles: Machine learning research experience developing predictive models, feature engineering pipelines and statistical validation workflows

6 Marko Skoric

Role: Partnerships Specialist Intern
Hometown: Tempe, AZ
Prior Relevant Roles: Treasurer at Sigma Nu Fraternity, M&A Analyst Intern at Sellside Group, LLC

7 Omoy Miller

Role: Growth & Operations Intern
Hometown: Philadelphia, PA
Qualified Impact Highlight: Led country diligence for Paraguay, assessing tariffs, constraints, and infrastructure feasibility

8 Shivane Rathod

Role: Strategic Finance Intern
Hometown: Monroe, NJ
Prior Relevant Roles: Wealth Mgmt. at Northwestern Mutual, Real Estate Intern at eXp Realty

9 Tammi Tran

Role: Marketing & Graphic Design
Hometown: Boston, MA
Qualified Impact Highlight: Front end @ Bread & Roses, Business Ops @ K36, & Usability Researcher @ Idea Lab

COMPANY ROADMAP
2026-2030

~~2025~~

Raised $500k upon creation, mobilized it in 2 months

Deployed 144 high efficiency crypto mining units in pilot Ethiopia site

Targeted initial daily output in USD ~$2,500

2026

Launch 5MW compute data center with government/PPP partnership

Expand into Nigeria, UAE & Finland

Prove heat reuse + community benefit partnerships

2027

Green Compute Park ~ 10–25 MW class

Cross border capacity, regional clients

Long-term off take / capacity reservation agreements

2028

50MW across sites, add redundancy

Development financing for scaled build outs and hyperscale

Structured product offerings

2029

Replicate the model into 2-3 frontier mkts.

Mining still revenue generator but AI/HPC center of growth

Bring SLAs, metering & Tier 1 reliability

2030

Most revenue from compute workloads

Strategic exit of crypto (via: large partner JV, buyout, or public readiness)

By 2029–2030, the plan is to replicate into multiple frontier markets, shift revenue toward diversified AI/HPC workloads with enterprise-grade SLAs, and position for a strategic exit via large buyout or listing on a capital markets.

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